Furong Huang

Associate Professor, University of Maryland

furongh@umd.edu • +1 301 405 8010 • furong-huang.com • 4124 Brendan Iribe Center

I certify that this curriculum vitae is a current and accurate statement of my professional record. Date: November 2, 2026

Signature: 

PERSONAL INFORMATION

Education

Doctor of Philosophy, Electrical and Computer Engineering
University of California, Irvine 09/2010-06/2016
Advisor: Anima Anandkumar

Bachelor of Science, Electrical Engineering and Computer Science
Zhejiang University, Zhejiang, China 09/2006-06/2010

Professional Experience

University of Maryland 07/2024-present
Associate Professor, Department of Computer Science (CS)
Associate Professor, Institute for Advanced Computer Studies (UMIACS)
Associate Professor, Maryland Robotics Center (MRC)
Affiliate, Applied Mathematics, Statistics, and Computation (AMSC)
Affiliate, Department of Electrical and Computer Engineering (ECE)

University of Maryland 07/2017-06/2024
Assistant Professor, Department of Computer Science (CS)
Assistant Professor, Institute for Advanced Computer Studies (UMIACS)
Affiliate, Applied Mathematics, Statistics, and Computation (AMSC)
Affiliate, Department of Electrical and Computer Engineering (ECE)

Microsoft Research, New York City 07/2016-07/2017
Postdoctoral Researcher
Mentors: John Langford, Robert Schapire

Microsoft Research, New England 06/2014-12/2014
Research Intern
Mentors: Jennifer Chayes, Christian Borgs

Microsoft Research, Cloud and Information Services Lab, Seattle 03/2014-05/2014
Research Intern
Mentors: Paul Mineiro, Nikos Karampatziakis

SCHOLARLY ACTIVITIES

Preprints and Manuscripts

P34 Yuan, Lingzhi, Yiguang Cai, Weize Liu, Siqi Yang, Feijiang Han, Yongkyun Lee, Udari Madhushani Sehwag, and Furong Huang. "Meta²: Precomputing Reusable Knowledge for Online Harness Evolution in Long-Horizon Agents." Paper. Project page, 2026.

P33 Lee, Jaewoo, Jeongyeon Seo, Sihyun Cho, Gyeongrak Choe, Yutong Wang, Bavin Saravanan, Jia-Bin Huang, Furong Huang, Sebastian Scherer, Guanya Shi, H. Jin Kim, Seungjae Lee, and Dongjae Lee. "Fly-by-Code: Embodied Coding Agents for Aerial Manipulation with Active Visual and Physical Feedback." arXiv. Project page, 2026.

P32 Lee, Jusuk, Sungha Kim, Yeonsoo Park, Jonguk Cheon, Yoonkyo Jung, Yongjun You, H. Jin Kim, Jia-Bin Huang, Furong Huang, Youngseok Jang, and Seungjae Lee. "Dex-One2Many: Learning Dexterous Manipulation from a Single Human Demonstration." arXiv. Project page, 2026.

P31 Shahidzadeh, Amir-Hossein, Seungjae Lee, Eadom Dessalene, Shanthosh Raaj Mohanram Mageswari, Soroush Etemad, Furong Huang, Cornelia Fermüller, and Yiannis Aloimonos. "What to Attend, What to Keep: Skill-Conditioned Visuotactile Representation with Progress-Guided Event Memory." arXiv, 2026.

P30 Liu, Minghui, Thomas Magelinski, Dehao Yuan, Qi Yu, and Furong Huang. "Ladders of Thought: A Self-Evolving Curriculum of Progressively Simplified Reasoning Traces." arXiv, 2026.

P29 Ho, Sy-Tuyen, Minghui Liu, Hieu Tran, and Furong Huang. "When AI Reviews Train AI Reviewers: Scientific-Judgment Collapse and Mitigation." arXiv, 2026.

P28 Yuan, Jiaxin, Udari Madhushani Sehwag, Connor Martinez Lockhart, Xiaoyu Liu, Shayan Shabihi, Jiaqi Wang, Chenghao Deng, Xiayimei Han, Vlassis Mastrantonis, Dmitrii Gudin, Shaopeng Zhu, Abdirisak Mohamed, Bilal Aytekin, Jiewen Lang, Zezheng Song, and Furong Huang. "MathAdv: What Theorem Provers Know, Reason, Formalize, and Generalize." arXiv. Code, 2026.

P27 Baskar, Sarvesh, Zikui Cai, Shayan Shabihi, Anirudh Satheesh, Muhammad R. Islam, Udari Madhushani Sehwag, Tom Goldstein, and Furong Huang. "The Low Frequency Trap: Video Language Models Fail at Simple Event Bookkeeping." arXiv, 2026.

P26 Xu, Jiawei, Minghui Liu, Juzheng Zhang, Tom Goldstein, and Furong Huang. "β-OPSD: Deriving with Policy Optimization, Training with Self-Distillation." arXiv, 2026.

P25 Pathmanathan, Pankayaraj, and Furong Huang. "Compliance2LoRA: Personalizable On-Demand Safety Alignment on Arbitrary Policy Subsets via Hypernetwork-Generated LoRA Adapters." arXiv, 2026.

P24 Liang, Yijun, Yunjie Tian, Yijiang Li, Yuqi Jia, Furong Huang, Tianyi Zhou, and Di Fu. "Visual Contrastive Self-Distillation." arXiv, 2026.

P23 Liu, Haowen, Xirui Li, Shaoxiong Yao, Peng Shi, Tianyi Zhou, Jia-Bin Huang, Furong Huang, and Jiayuan Mao. "Guava: Distilling Frontier VLMs into a Compact Agent through a Robotic Manipulation Harness." arXiv, 2026.

P22 Liu, Sida, Feijiang Han, Sarah Wiegreffe, and Furong Huang. "Non-Gaussianity Reveals Interpretable Directions in Language Models." arXiv. Project page. Code, 2026.

P21 Agrawal, Aakriti, Souradip Chakraborty, Armin Saghafian, Nihal Sharma, Rizal Fathony, Nam H. Nguyen, C. Bayan Bruss, Amrit Singh Bedi, and Furong Huang. "The Hidden Bias of Process Reward Models: PRISM for Rewarding the Right Reasoning." arXiv, 2026.

P20 Agrawal, Aakriti, Minghui Liu, and Furong Huang. "VeriGate: Verifier-Gated Step-Level Supervision for GRPO." arXiv, 2026.

P19 Ho, Sy-Tuyen, Minghui Liu, Huy Nghiem, and Furong Huang. "SoundnessBench: Can Your AI Scientist Really Tell Good Research Ideas from Bad Ones?" arXiv, 2026.

P18 Ankireddy, Sravan Kumar, Nikita Seleznev, Nam H. Nguyen, Yulun Wu, Senthil Kumar, Furong Huang, and C. Bayan Bruss. "TimeSqueeze: Dynamic Patching for Efficient Time Series Forecasting." arXiv, 2026.

P17 Liu, Weize, Minghui Liu, Sy-Tuyen Ho, Souradip Chakraborty, Xiyao Wang, and Furong Huang. "Agentic Critical Training." arXiv. Project page, 2026.

P16 Stein, Alex, Furong Huang, and Tom Goldstein. "GATES: Self-Distillation under Privileged Context with Consensus Gating." arXiv, 2026.

P15 Satheesh, Anirudh, Ziyi Chen, Furong Huang, and Heng Huang. "Provably Efficient Algorithms for S- and Non-Rectangular Robust MDPs with General Parameterization." arXiv, 2026.

P14 Yuan, Dehao, Tyler Farnan, Stefan Tesliuc, Doron L. Bergman, Yulun Wu, Xiaoyu Liu, Minghui Liu, James Montgomery, Nam H. Nguyen, C. Bayan Bruss, and Furong Huang. "PersonaLedger: Generating Realistic Financial Transactions with Persona Conditioned LLMs and Rule Grounded Feedback." arXiv, 2026.

P13 Pathmanathan, Pankayaraj, Michael-Andrei Panaitescu-Liess, Cho-Yu Jason Chiang, and Furong Huang. "RAGPart & RAGMask: Retrieval-Stage Defenses Against Corpus Poisoning in Retrieval-Augmented Generation." arXiv, 2025.

P12 Liu, Minghui, Aadi Palnitkar, Tahseen Rabbani, Hyunwoo Jae, Kyle Rui Sang, Dixi Yao, Shayan Shabihi, Fuheng Zhao, Tian Li, Ce Zhang, Furong Huang, and Kunpeng Zhang. "Hold Onto That Thought: Assessing KV Cache Compression on Reasoning." arXiv, 2025.

P11 Zhai, Kevin, Utsav Singh, Anirudh Thatipelli, Souradip Chakraborty, Anit Kumar Sahu, Furong Huang, Amrit Singh Bedi, and Mubarak Shah. "MIRA: Towards Mitigating Reward Hacking in Inference-Time Alignment of T2I Diffusion Models." arXiv, 2025.

P10 Wang, Xiyao, Chunyuan Li, Jianwei Yang, Kai Zhang, Bo Liu, Tianyi Xiong, and Furong Huang. "LLaVA-Critic-R1: Your Critic Model Is Secretly a Strong Policy Model." arXiv, 2025.

P9 Cai, Zikui, Andrew Wang, Anirudh Satheesh, Ankit Nakhawa, Hyunwoo Jae, Keenan Powell, Minghui Liu, Neel Jay, Sungbin Oh, Xiyao Wang, Yongyuan Liang, Tom Goldstein, and Furong Huang. "MORSE-500: A Programmatically Controllable Video Benchmark to Stress-Test Multimodal Reasoning." arXiv, 2025.

P8 Chakraborty, Souradip, Mohammadreza Pourreza, Ruoxi Sun, Yiwen Song, Nino Scherrer, Furong Huang, Amrit Singh Bedi, Ahmad Beirami, Jindong Gu, Hamid Palangi, and Tomas Pfister. "On the Role of Feedback in Test-Time Scaling

of Agentic AI Workflows." arXiv, 2025.

P7 Liu, Minghui, Tahseen Rabbani, Tony O'Halloran, Ananth Sankaralingam, Mary-Anne Hartley, Furong Huang, Cornelia Fermüller, and Yiannis Aloimonos. "HashEvict: A Pre-Attention KV Cache Eviction Strategy using Locality-Sensitive Hashing." arXiv, 2024.

P6 Li, Lincan, Jiaqi Li, Catherine Chen, Fred Gui, Hongjia Yang, Chenxiao Yu, Zhengguang Wang, Jianing Cai, Junlong Aaron Zhou, Bolin Shen, Alex Qian, Weixin Chen, Zhongkai Xue, Lichao Sun, Lifang He, Hanjie Chen, Kaize Ding, Zijian Du, Fangzhou Mu, Jiaxin Pei, Jieyu Zhao, Swabha Swayamdipta, Willie Neiswanger, Hua Wei, Xiyang Hu, Shixiang Zhu, Tianlong Chen, Yingzhou Lu, Yang Shi, Lianhui Qin, Tianfan Fu, Zhengzhong Tu, Yuzhe Yang, Jaemin Yoo, Jiaheng Zhang, Ryan Rossi, Liang Zhan, Liang Zhao, Emilio Ferrara, Yan Liu, Furong Huang, Xiangliang Zhang, Lawrence Rothenberg, Shuiwang Ji, Philip S. Yu, Yue Zhao, and Yushun Dong. "Political-LLM: Large Language Models in Political Science." arXiv, 2024.

P5 Wang, Xiyao, Linfeng Song, Ye Tian, Dian Yu, Baolin Peng, Haitao Mi, Furong Huang, and Dong Yu. "Towards Self-Improvement of LLMs via MCTS: Leveraging Stepwise Knowledge with Curriculum Preference Learning." arXiv, 2024.

P4 Ding, Mucong, Yuancheng Xu, Tahseen Rabbani, Xiaoyu Liu, Brian Gravelle, Teresa Ranadive, Tai-Ching Tuan, and Furong Huang. "Calibrated Dataset Condensation for Faster Hyperparameter Search." arXiv, 2024.

P3 Ding, Mucong, Yinhan He, Jundong Li, and Furong Huang. "Spectral Greedy Coresets for Graph Neural Networks." arXiv, 2024.

P2 Rabbani, Tahseen, Jiahao Su, Xiaoyu Liu, David Chan, Geoffrey Sangston, and Furong Huang. "conv_einsum: A Framework for Representation and Fast Evaluation of Multilinear Operations in Convolutional Tensorial Neural Networks." arXiv, 2024.

P1 Gitter, Anthony, Furong Huang, Ragupathyraj Valluvan, Ernest Fraenkel, and Animashree Anandkumar. "Unsupervised learning of transcriptional regulatory networks via latent tree graphical models." arXiv, 2016.

Conference Publications

138 items

C138 Chen, Yuen, Yulun Wu, Samuel Sharpe, Igor Melnyk, Nam H. Nguyen, Furong Huang, C. Bayan Bruss, and Rizal Fathony. "Why Freeze the Encoder? A Controlled Study of End-to-End Sequence–Graph Learning for User Behavior." arXiv. OpenReview. Learning on Graphs Conference (LoG), Poster, 2026.

C137 Yuan, Lingzhi, Chenghao Deng, Fangxu Yu, Souradip Chakraborty, Mohammad Rostami, and Furong Huang. "FlowBank: Query-Adaptive Agentic Workflows Optimization through Precompute-and-Reuse." arXiv. Conference on Neural Information Processing Systems (NeurIPS), Main Track, Poster, 2026.

C136 Lee, Jusuk, Seungjae Lee, Jonghun Shin, Hoseong Jung, Sungha Kim, Daesol Cho, H. Jin Kim, Jia-Bin Huang, and Furong Huang. "DynaFLIP: Rethinking Robotics Perception via Tri-Modal-Dynamics Guided Representation." arXiv. Conference on Neural Information Processing Systems (NeurIPS), Main Track, Poster, 2026.

C135 Lee, Seungjae, Yoonkyo Jung, Jusuk Lee, Jonghun Shin, Amir Hossein Shahidzadeh, Yao-Chih Lee, H. Jin Kim, Jia-Bin Huang, and Furong Huang. "μ0: A Scalable 3D Interaction-Trace World Model." arXiv. OpenReview. Project page. Code. Models. Conference on Robot Learning (CoRL), Poster, 2026.

C134 Zheng, Ruijie, Dantong Niu, Yuqi Xie, Jing Wang, Mengda Xu, Yunfan Jiang, Fernando Castañeda, Fengyuan Hu, You Liang Tan, Letian Fu, Trevor Darrell, Furong Huang, Yuke Zhu, Danfei Xu, and Linxi Fan. "EgoScale: Scaling Dexterous Manipulation with Diverse Egocentric Human Data." arXiv. OpenReview. Conference on Robot Learning (CoRL), Oral, 2026.

C133 Kini, Prajakta, Avinash Reddy, Souradip Chakraborty, Satya Sai Srinath Namburi GNVV, Furong Huang, Amrit Singh Bedi, and Alvaro Velasquez. "Does Reasoning Preserve Alignment? On the Trustworthiness of Large Reasoning Models." arXiv. Findings of the Association for Computational Linguistics: EMNLP, 2026.

C132 Cai, Zikui, Kaushal Janga, Tan Dat Dao, Seungjae Lee, Shivin Dass, Mingyo Seo, Kaiyu Yue, Mintong Kang, Nandhu Pillai, Monte Hoover, Aadi Palnitkar, Ruchit Rawal, Ruijie Zheng, Bo Li, Yuke Zhu, Roberto Martín-Martín, Tom Goldstein, and Furong Huang. "Beyond Episodic Evaluation: Memory Architectural Bottlenecks in Sequential Embodied Question Answering." arXiv. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.

C131 Xu, Jiawei, Minghui Liu, Aakriti Agrawal, Yifan Chen, and Furong Huang. "Scheduling Thoughts: Learning the Order of Thought in Diffusion Language Models." arXiv. openreview. Forty-third International Conference on Machine Learning (ICML), 2026.

C130 Zheng, Tong, Chengsong Huang, Runpeng Dai, Yun He, Rui Liu, Xin Ni, Huiwen Bao, Kaishen Wang, Hongtu Zhu, Jiaxin Huang, Furong Huang, and Heng Huang. "Parallel-Probe: Towards Efficient Parallel Thinking via 2D Probing." arXiv. Forty-third International Conference on Machine Learning (ICML), 2026.

C129 Yu, Fangxu, Xingang Guo, Lingzhi Yuan, Haoqiang Kang, Hongyu Zhao, Lianhui Qin, Furong Huang, Bin Hu, and Tianyi Zhou. "TSRBench: A Comprehensive Multi-task Multi-modal Time Series Reasoning Benchmark for Generalist Models." arXiv. Project page. Forty-third International Conference on Machine Learning (ICML), 2026.

C128 Lazri, Zachary McBride, Anirudh Nakra, Ivan Brugere, Danial Dervovic, Antigoni Polychroniadou, Furong Huang, Dana Dachman-Soled, and Min Wu. "MAFE: Enabling Equitable Algorithm Design in Multi-Agent Multi-Stage Decision-Making Systems." arXiv. Forty-third International Conference on Machine Learning (ICML), 2026.

C127 Ghosal, Soumya Suvra, Souradip Chakraborty, Vaibhav Singh, Furong Huang, Dinesh Manocha, and Amrit Singh Bedi. "Safety Recovery in Reasoning Models Is Only a Few Early Steering Steps Away." arXiv. Forty-third International Conference on Machine Learning (ICML), 2026.

C126 Xiong, Nuoya, Yuhang Zhou, Hanqing Zeng, Zhaorun Chen, Furong Huang, Shuchao Bi, Lizhu Zhang, and Zhuokai Zhao. "Token-Level LLM Collaboration via FusionRoute." arXiv. Forty-third International Conference on Machine Learning (ICML), 2026.

C125 Sehwag, Udari Madhushani, Elaine Lau, Haniyeh Ehsani Oskouie, Shayan Shabihi, Erich Liang, Andrea Sarai Echeverria Toledo, Guillermo A. Mangialardi, Sergio Fonrouge, Ed-Yeremai Hernandez-Cardona, Paula Vergara, Utkarsh Tyagi, Chen Bo Calvin Zhang, Pavi Bhatter, Nicholas E. Johnson, Furong Huang, Ernesto Gabriel Hernández Montoya, and Bing Liu. "SciPredict: Can LLMs Predict the Outcomes of Scientific Experiments in Natural Sciences?" arXiv. Code. Forty-third International Conference on Machine Learning (ICML), 2026.

C124 Bornstein, Marco, Zora Che, Suhas Julapalli, Abdirisak Mohamed, Amrit Singh Bedi, and Furong Huang. "Advancing Regulation in Artificial Intelligence: An Auction-Based Approach." arXiv. The Ninth Annual ACM Conference on Fairness, Accountability, and Transparency (FAccT), 2026.

C123 Pathmanathan, Pankayaraj, and Furong Huang. "Teach a Reward Model to Correct Itself: Reward Guided Adversarial Failure Discovery for Robust Reward Modeling." arXiv. Code. Main Conference, The 64th Annual Meeting of the Association for Computational Linguistics (ACL), **Oral**, 2026.

C122 Agrawal, Aakriti, Gouthaman KV, Rohith Aralikatti, Gauri Jagatap, Jiaxin Yuan, Sarvesh Baskar, Vijay Kamarshi, Andrea Fanelli, and Furong Huang. "Towards Mitigating Hallucinations in Large Vision-Language Models by Refining Textual Embeddings." arXiv. Findings, The 64th Annual Meeting of the Association for Computational Linguistics (ACL), 2026.

C121 Agrawal, Aakriti, Mucong Ding, Chenghao Deng, Zora Che, Arjun Rajaram, Anirudh Satheesh, Bang An, C. Bayan Bruss, John Langford, and Furong Huang. "EnsemW2S: Enhancing Weak-to-Strong Generalization with Large Language Model Ensembles." arXiv. Proceedings. DOI. Findings, The 64th Annual Meeting of the Association for Computational Linguistics (ACL), 2026.

C120 Lee, Seungjae, Yoonkyo Jung, Inkook Chun, Yao-Chih Lee, Zikui Cai, Hongjia Huang, Aayush Talreja, Tan Dat Dao, Yongyuan Liang, Jia-Bin Huang and Furong Huang. "TraceGen: World Modeling in 3D Trace Space Enables Learning from Cross-Embodiment Videos." arXiv. Project page. Code. Conference on Computer Vision and Pattern Recognition (CVPR), 2026.

C119 Ju, Yuanchen, Yongyuan Liang, Yen-Jen Wang, Nandiraju Gireesh, Yuanliang Ju, Seungjae Lee, Qiao Gu, Elvis Hsieh, Furong Huang*, and Koushil Sreenath*. "MomaGraph: State-Aware Unified Scene Graphs with Vision-Language Model for Embodied Task Planning." arXiv. Project page. Code. The Fourteenth International Conference on Learning Representations (ICLR), **Oral**, 2026.

C118 Liang, Yongyuan, Wei Chow, Feng Li, Ziqiao Ma, Xiyao Wang, Jiageng Mao, Jiuhai Chen, Jiatao Gu, Yue Wang, and Furong Huang. "ROVER: Benchmarking Reciprocal Cross-Modal Reasoning for Omnimodal Generation." arXiv. Project page. Code. The Fourteenth International Conference on Learning Representations (ICLR), 2026.

C117 Sehwag, Udari Madhushani, Shayan Shabihi, Alex McAvoy, Vikash Sehwag, Yuancheng Xu, Dalton Towers, and Furong Huang. "PropensityBench: Evaluating Latent Safety Risks in Large Language Models via an Agentic Approach." arXiv. Code. The Fourteenth International Conference on Learning Representations (ICLR), 2026.

C116 Li, Ang, Charles L. Wang, Deqing Fu, Kaiyu Yue, Zikui Cai, Wang Bill Zhu, Ollie Liu, Peng Guo, Willie Neiswanger, Furong Huang, Tom Goldstein and Micah Goldblum. "Zebra-CoT: A Dataset for Interleaved Vision-Language Reasoning." arXiv. Dataset. The Fourteenth International Conference on Learning Representations (ICLR), 2026.

C115 Huang, Yue, Chujie Gao, Siyuan Wu, Haoran Wang, Xiangqi Wang, Jiayi Ye, Yujun Zhou, Yanbo Wang, Jiawen Shi, Qihui Zhang, Han Bao, Zhaoyi Liu, Yuan Li, Tianrui Guan, Peiran Wang, Haomin Zhuang, Dongping Chen, Kehan Guo, Andy Zou, Bryan Hooi, Caiming Xiong, Elias Stengel-Eskin, Hongyang Zhang, Hongzhi Yin, Huan Zhang, Huaxiu Yao, Jieyu Zhang, Jaehong Yoon, Kai Shu, Ranjay Krishna, Swabha Swayamdipta, Weijia Shi, Xiang Li, Yuexing Hao, Zhihao Jia, Zhize Li, Xiuying Chen, Zhengzhong Tu, Xiyang Hu, Tianyi Zhou, Jieyu Zhao, Lichao Sun, Furong Huang, Or Cohen-Sasson, Prasanna Sattigeri, Anka Reuel, Max Lamparth, Yue Zhao, Nouha Dziri, Yu Su, Huan Sun, Heng Ji, Chaowei Xiao, Mohit Bansal, Nitesh V Chawla, Jian Pei, Jianfeng Gao, Michael Backes, Philip S. Yu, Neil Zhenqiang Gong, Pin-Yu Chen, Bo Li, Dawn Song, Xiangliang Zhang. "TrustGen: A Platform of Dynamic Benchmarking on the Trustworthiness of Generative Foundation Models." arXiv. The Fourteenth International Conference on Learning Representations (ICLR), 2026.

C114 Pathmanathan, Pankayaraj, Udari Madhushani Sehwag, Michael-Andrei Panaitescu-Liess, Cho-Yu Jason Chiang and Furong Huang. "AdvBDGen: A Robust Framework for Generating Adaptive and Stealthy Backdoors in LLM Alignment Attacks." arXiv. Code. Project page. AAAI 2026 AI Alignment Track (AAAI), **Oral**, 2026.

C113 Beetham, James, Souradip Chakraborty, Mengdi Wang, Furong Huang, Amrit Singh Bedi and Mubarak Shah. "Jailbreaks as Inference-Time Alignment: A Framework for Understanding Safety Failures in LLMs." paper. 19th Conference of the

European Chapter of the Association for Computational Linguistics (EACL), 2026.

C112 Wang, Xiyao, Zhengyuan Yang, Chao Feng, Hongjin Lu, Linjie Li, Chung-Ching Lin, Kevin Lin, Furong Huang*, and Lijuan Wang*. "SoTA with Less: MCTS-Guided Sample Selection for Data-Efficient Visual Reasoning Self-Improvement." arXiv. Code. The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS), **Spotlight**, 2025.

C111 Nguyen, Tin Trung, Jiannan Xu, Zora Che, Phuong-Anh Nguyen-Le, Rushil Dandamudi, Donald Braman, Furong Huang, Hal Daumé III, and Zubin Jelveh. "Effort-aware Fairness: Incorporating a Philosophy-informed, Human-centered Notion of Effort into Algorithmic Fairness Metrics." Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (AIES), 8(2):1875–1887, 2025. arXiv. doi.

C110 Wang, Xiyao, Zhengyuan Yang, Chao Feng, Yuhang Zhou, Xiaoyu Liu, Yongyuan Liang, Ming Li, Ziyi Zang, Linjie Li, Chung-Ching Lin, Kevin Lin, Furong Huang* and Lijuan Wang*. "ViCrit: A Verifiable Reinforcement Learning Proxy Task for Visual Perception in VLMs." arXiv. Code. The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS), 2025.

C109 Ghosal, Soumya Suvra, Souradip Chakraborty, Avinash Reddy, Yifu Lu, Mengdi Wang, Dinesh Manocha, Furong Huang, Mohammad Ghavamzadeh and Amrit Singh Bedi. "Does Thinking More Always Help? Mirage of Test-Time Scaling in Reasoning Models." arXiv. The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS), 2025.

C108 Ding, Mucong, Bang An, Tahseen Rabbani, Chenghao Deng, Anirudh Satheesh, Souradip Chakraborty, Mehrdad Saberi, Yuxin Wen, Kyle Rui Sang, Aakriti Agrawal, Xuandong Zhao, Mo Zhou, Mary-Anne Hartley, Lei Li, Yu-Xiang Wang, Vishal M. Patel, Soheil Feizi, Tom Goldstein and Furong Huang. "A Technical Report on "Erasing the Invisible": The 2024 NeurIPS Competition on Stress Testing Image Watermarks." openreview. Competition website (beige box). Competition website (black box). Project page. The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track (NeurIPS), 2025.

C107 Agrawal, Aakriti, Rohith Aralikatti, Anirudh Satheesh, Souradip Chakraborty, Amrit Singh Bedi, Furong Huang. "Uncertainty-Aware Answer Selection for Improved Reasoning in Multi-LLM Systems." arXiv. The 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025.

C106 Zhou, Yuhang, Jing Zhu, Shengyi Qian, Zhuokai Zhao, Xiyao Wang, Xiaoyu Liu, Ming Li, Paiheng Xu, Wei Ai, Furong Huang. "DISCO Balances the Scales: Adaptive Domain- and Difficulty-Aware Reinforcement Learning on Imbalanced Data." arXiv. Code. The 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025.

C105 Lee, Seungjae[†], Daniel Ekpo[†], Haowen Liu, Furong Huang[*], Abhinav Shrivastava[*], Jia-Bin Huang[*]. "Imagine, Verify, Execute: Agentic Exploration with Vision-Language Models." arXiv. Code. Project page. 9th Annual Conference on Robot Learning (CoRL), 2025.

C104 Zheng, Ruijie, Jing Wang, Scott Reed, Yu Fang, Fengyuan Hu, Joel Jang, Kaushil Kundalia, Zongyu Lin, Loïc Magne, Avnish Narayan, You Liang Tan, Guanzhi Wang, Qi Wang, Jiannan Xiang, Yinzhen Xu, Seonghyeon Ye, Jan Kautz, Furong Huang, Yuke Zhu, Linxi Fan. "FLARE: Robot Learning with Implicit World Modeling." arXiv. 9th Annual Conference on Robot Learning (CoRL), 2025.

C103 Yu, Kelin, Sheng Zhang, Harshit Soora, Furong Huang, Heng Huang, Pratap Tokekar, and Ruohan Gao. "GenFlowRL: Shaping Rewards with Generative Object-Centric Flow in Visual Reinforcement Learning." arXiv. Project page. International Conference on Computer Vision (ICCV), 2025.

C102 Wang, Xiyao, Zhengyuan Yang, Linjie Li, Hongjin Lu, Yuancheng Xu, Chung-Ching Lin, Kevin Lin, Furong Huang[*], and Lijuan Wang[*]. "VisVM: Scaling Inference-time Search with Vision Value Model for Improved Visual Comprehension." arXiv. Code. Project page. International Conference on Computer Vision (ICCV), 2025.

C101 Yue, Kaiyu, Vasu Singla, Menglin Jia, John Kirchenbauer, Rifaa Qadri, Zikui Cai, Abhinav Bhatele, Furong Huang, and Tom Goldstein. "Zero-Shot Vision Encoder Grafting via LLM Surrogates." arXiv. International Conference on Computer Vision (ICCV), 2025.

C100 Joshua McClellan, Greyson Brothers, Furong Huang, and Pratap Tokekar. "PEnGUiN: Partially Equivariant Graph NeUral Networks for Sample Efficient MARL." arXiv. Reinforcement Learning Conference (RLC), 2025.

C99 Soumya Suvra Ghosal, Souradip Chakraborty, Vaibhav Singh, Tianrui Guan, Mengdi Wang, Ahmad Beirami, Furong Huang, Alvaro Velasquez, Dinesh Manocha, and Amrit Singh Bedi. "Immune: Improving Safety Against Jailbreaks in Multi-modal LLMs via Inference-Time Alignment." arXiv. Conference on Computer Vision and Pattern Recognition (CVPR), 2025.

C98 Zheng, Ruijie, Yongyuan Liang, Shuaiyi Huang, Jianfeng Gao, Hal Daume III, Andrey Kolobov, Furong Huang, and Jianwei Yang. "TraceVLA: Visual Trace Prompting Enhances Spatial-Temporal Awareness for Generalist Robotic Policies." arXiv. Code. Project page. The Thirteenth International Conference on Learning Representations (ICLR), 2025.

C97 Xu, Yuancheng, Udari Madhushani Sehwag, Alec Koppel, Sicheng Zhu, Bang An, Furong Huang, and Sumitra Ganesh. "GenARM: Reward Guided Generation with Autoregressive Reward Model for Test-Time Alignment." arXiv. Project page. Code. The Thirteenth International Conference on Learning Representations (ICLR), 2025.

C96 Yingzi Ma, Jiongxiao Wang, Fei Wang, Siyuan Ma, Jiazhao Li, Jinsheng Pan, Xiujun Li, Furong Huang, Lichao Sun,

Bo Li, Yejin Choi, Muhao Chen, and Chaowei Xiao. "Benchmarking Vision Language Model Unlearning via Fictitious Facial Identity Dataset." arXiv. Project page. The Thirteenth International Conference on Learning Representations (ICLR), 2025.

C95 Chakraborty, Souradip, Sujay Bhatt, Udari Madhushani Sehwag, Soumya Suvra Ghosal, Jiahao Qiu, Mengdi Wang, Dinesh Manocha, Furong Huang, Alec Koppel, and Sumitra Ganesh. "Collab: Controlled Decoding using Mixture of Agents for LLM Alignment." arXiv. Project page. The Thirteenth International Conference on Learning Representations (ICLR), 2025.

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J6 Su, Jiahao, Jingling Li, Xiaoyu Liu, Teresa Ranadive, Christopher Coley, Tai-Ching Tuan, and Furong Huang. "Compact Neural Architecture Designs by Tensor Representations." Frontiers in Artificial Intelligence-Machine Learning and Artificial Intelligence, 2022.

J5 Giraud, Luc, Ulrich Rüde, Linda Stals, Paolo Bientinesi, David Ham, Furong Huang, Paul HJ Kelly, Christian Lengauer, and Saday Sadayappan. "Dagstuhl Reports, Vol. 10, Issue 3 ISSN 2192-5283." (2020).

J4 Huang, Furong, and Animashree Anandkumar. "Convolutional Dictionary Learning through Tensor Factorization." Journal of Machine Learning Research conference and workshop proceedings 2015.

J3 Huang, Furong, U. N. Niranjan, Mohammad Umar Hakeem, and Animashree Anandkumar. "Fast Detection of Overlapping Communities via Online Tensor Methods/ Online Tensor Methods for Learning Latent Variable Models." Journal of Machine Learning Research 2014.

J2 Anandkumar, Animashree, Vincent YF Tan, Furong Huang, and Alan S. Willsky. "High-Dimensional Gaussian Graphical Model Selection: Walk-Summability and Local Separation Criterion." Journal of Machine Learning Research, Aug. 2012. An abridged version appears in the Conference on Neural Information Processing Systems, Dec. 2011.

J1 Anandkumar, Animashree, Vincent YF Tan, Furong Huang, and Alan S. Willsky. "High-Dimensional Structure Learning of Ising Models: Local Separation Criterion." Annals of Statistics, Volume 40, Number 3 (2012), 1346-1375. An abridged version appears in the Proc. of NIPS, Dec. 2011.

Refereed Workshop Publications

106 items

W106 Li, Jie S., Kwesi Adu Cobbina, Udari Madhushani Sehwag, David Yu Miller, Hirunima Jayasekara, Shayan Shabihi, Zikui Cai, Furong Huang, and Tom Goldstein. "CARVE: Caption-Anchored, Rubric-Verified Extraction for Training-Scale Figure-Required Scientific VQA." VLM4RWD Workshop: Grounded and Faithful Vision-Language Models for Real-World Deployment, poster, NeurIPS 2026. OpenReview.

W105 Liu, Haowen, Xirui Li, Shaoxiong Yao, Peng Shi, Tianyi Zhou, Jia-Bin Huang, Furong Huang, and Jiayuan Mao. "Guava: Distilling Frontier VLMs into a Compact Agent through a Robotic Manipulation Harness." 4th Workshop on Learning Effective Abstractions for Planning (LEAP), CoRL 2026. OpenReview. arXiv.

W104 Liu, Haowen, Xirui Li, Shaoxiong Yao, Peng Shi, Tianyi Zhou, Jia-Bin Huang, Furong Huang, and Jiayuan Mao. "Guava: Distilling Frontier VLMs into a Compact Agent through a Robotic Manipulation Harness." Workshop on Agentic Robotics, poster, CoRL 2026. OpenReview. arXiv.

W103 Lee, Jusuk, Sungha Kim, Yeonsoo Park, Jonguk Cheon, Yoonkyo Jung, Yongjun You, H. Jin Kim, Jia-Bin Huang, Furong Huang, Youngseok Jang, and Seungjae Lee. "Dex-One2Many: Learning Dexterous Manipulation from a Single Human Demonstration." Scaling H2R Workshop, poster and oral, CoRL 2026. OpenReview. arXiv.

W102 Shahidzadeh, Amir-Hossein, Seungjae Lee, Eadom T. Dessalene, Shanthosh Raaj Mohanram Mageswari, Soroush Etemad, Furong Huang, Cornelia Fermüller, and Yiannis Aloimonos. "What to Attend, What to Keep: Skill-Conditioned Visuotactile Representation with Progress-Guided Event Memory." Scaling H2R Workshop, poster, CoRL 2026. OpenReview. arXiv.

W101 Shahidzadeh, Amir-Hossein, Seungjae Lee, Eadom Dessalene, Shanthosh Raaj Mohanram Mageswari, Soroush Etemad, Furong Huang, Cornelia Fermüller, and Yiannis Aloimonos. "What to Attend, What to Keep: Skill-Conditioned Visuotactile Representation with Progress-Guided Event Memory." Touch2Action Workshop, IROS 2026. **Best Paper Award**. arXiv. Project page.

W100 Pathmanathan, Pankayaraj, and Furong Huang. "Compliance2LoRA: On-Demand Safety Alignment on Arbitrary Policy Subsets via Hypernetwork-Generated LoRA Adapters." BiAlign Workshop, poster, NeurIPS 2026. OpenReview. arXiv.

W99 Lee, Jusuk, Seungjae Lee, Jonghun Shin, Hoseong Jung, Sungha Kim, Daesol Cho, H. Jin Kim, Jia-Bin Huang, and Furong Huang. "DynaFLIP: Rethinking Robotics Perception via Tri-Modal-Dynamics Guided Representation." Robotics World Modeling Workshop, NeurIPS 2026. OpenReview. arXiv.

W98 Ho, Sy-Tuyen, Anirudh Satheesh, Soham Chawathe, Huy Nghiem, Yongkyun Lee, Andrew Adams Schoen, and Furong Huang. "SpectralCanary: Embarrassingly Simple Low-Frequency Traces for T2I Dataset-Use Auditing." ATTRIB Workshop, NeurIPS 2026. OpenReview.

W97 Ho, Sy-Tuyen, Minghui Liu, and Furong Huang. "Recursive Training and Scientific Judgment in AI Peer Review." FLLMPT Workshop, poster, NeurIPS 2026. OpenReview. arXiv.

W96 Pathmanathan, Pankayaraj, and Furong Huang. "Compliance2LoRA: On-Demand Safety Alignment on Arbitrary Policy Subsets via Hypernetwork-Generated LoRA Adapters." TAE (Trust-AI-Eval): Can We Trust AI Evaluation?, poster, non-archival, NeurIPS 2026. OpenReview. arXiv.

W95 Wang, Zhi, Botao He, Kelin Yu, Seungjae Lee, Ruohan Gao, Furong Huang, and Yiannis Aloimonos. "HumanEgo: A Zero-Instrumentation Data-to-Deployment Pipeline for Robot Learning from Human Egocentric Video." ScaleInfra Workshop, oral, IROS 2026. **Best Demo Paper Award**. OpenReview. arXiv.

W94 Wang, Zhi, Botao He, Kelin Yu, Seungjae Lee, Ruohan Gao, Furong Huang, and Yiannis Aloimonos. "Spatial Intelligence and Predictive Objectives for Zero-Shot Manipulation from Minutes of Egocentric Video." WORLDS: World Models and Spatial Intelligence for Physical AI, poster and spotlight, IROS 2026. **Best Paper Award**. OpenReview. arXiv.

W93 Wang, Zhi, Botao He, Kelin Yu, Seungjae Lee, Ruohan Gao, Furong Huang, and Yiannis Aloimonos. "HumanEgo: Zero-Shot Robot Learning from Minutes of Human Egocentric Videos." Scaling H2R Workshop, poster, CoRL 2026. OpenReview. arXiv.

W92 Lee, Seungjae, Yoonkyo Jung, Jusuk Lee, Jonghun Shin, Amir Hossein Shahidzadeh, Yao-Chih Lee, H. Jin Kim, Jia-Bin Huang, and Furong Huang. "μ_0: A Scalable 3D Interaction-Trace World Model." RoBoWoMo Workshop, poster and spotlight, IROS 2026. **Second-place Best Paper Award**. OpenReview. arXiv.

W91 Pathmanathan,Pankayaraj, and Furong Huang. "Deliberative Alignment is Deep, but Uncertainty Remains: Inference time safety improvement in reasoning via attribution of unsafe behavior to base model." Actionable Interpretability Workshop (AIW), COLM 2026. arXiv.

W90 Wang, Zhi, Botao He, Kelin Yu, Seungjae Lee, Ruohan Gao, Furong Huang, Yiannis Aloimonos. "Interaction-Centric Tokens: A Representation for Cross-Embodiment Manipulation." Workshop on From Perception to Action: Representation-Centric Robot Autonomy, RSS 2026.

W89 Wang, Zhi, Botao He, Kelin Yu, Seungjae Lee, Ruohan Gao, Furong Huang, and Yiannis Aloimonos. "HumanEgo: Zero-Shot Robot Learning from Minutes of Human Egocentric Videos." Workshop on Data-Centric Robotics: What Data Do Robots Really Need, RSS 2026. arXiv.

W88 Satheesh, Anirudh, Michael-Andrei Panaitescu-Liess, Andrew Ye Xu, Georgios Milis, Heng Huang, Zikui Cai, and Furong Huang. "Compositional Adversarial Training for Robust Visual Watermarking." 2nd Workshop on Compositional Learning: Safety, Interpretability, and Agents, ICML 2026. arXiv.

W87 Panaitescu-Liess, Michael-Andrei, Yigitcan Kaya, Fiza Mulla, Anirudh Satheesh, and Furong Huang. "When Detection Tools Collide: On the Impact of Watermarking on AI-Generated Text Detectors." 4th Workshop on Towards Knowledgeable Foundation Models (KnowFM), ACL 2026.

W86 Panaitescu-Liess, Michael-Andrei, Aadi Palnitkar, Archit Kambhamettu, Yigitcan Kaya, Daniel Brown, Sungbin Oh, Sean Michael McLeish, Marco Bornstein, Furong Huang, and Tom Goldstein. "Practical Memorization Tests for Detecting Copyrighted Data in Large Language Models." Seventh Workshop on Privacy in Natural Language Processing (PrivateNLP), ACL 2026.

W85 Bornstein, Marco, Zora Che, Suhas Julapalli, Abdirisak Mohamed, Amrit Singh Bedi, and Furong Huang. "Advancing Regulation in Artificial Intelligence: An Auction-Based Approach." Workshop on AI for Mechanism Design and Strategic Decision Making (AIMS), ICLR 2026. OpenReview. arXiv.

W84 Pathmanathan, Pankayaraj, Michael-Andrei Panaitescu-Liess, Cho-Yu Jason Chiang, and Furong Huang. "RAGPart & RAGMask: Retrieval-Stage Defenses Against Corpus Poisoning in Retrieval-Augmented Generation." FrontierIR Workshop, oral, non-archival, AAAI 2026. OpenReview. arXiv.

W83 Pathmanathan, Pankayaraj, and Furong Huang. "Teach a Reward Model to Correct Itself: Reward Guided Adversarial Failure Discovery for Robust Reward Modeling." Workshop on Trust and Control in Agentic AI (TrustAgent), AAAI 2026. OpenReview. arXiv.

W82 Chen, Yuen, Yulun Wu, Samuel Sharpe, Igor Melnyk, Nam H. Nguyen, Furong Huang, C. Bayan Bruss, and Rizal Fathony. "Bridging the Divide: End-to-End Sequence-Graph Learning." Workshop on New Perspective in Graph Machine Learning (NPGML), NeurIPS 2025. OpenReview. arXiv.

W81 Ankireddy, Sravan Kumar, Nikita Seleznev, Nam H. Nguyen, Yulun Wu, Senthil Kumar, Furong Huang, and C. Bayan Bruss. "TimeSqueeze: Dynamic Patching for Efficient Time Series Forecasting." BERT^{2}S Workshop, NeurIPS 2025.

OpenReview. arXiv.

W80 Liu, Minghui, Aadi Palnitkar, Tahseen Rabbani, Hyunwoo Jae, Kyle Rui Sang, Dixi Yao, Shayan Shabihi, Fuheng Zhao, Tian Li, Ce Zhang, Furong Huang, and Kunpeng Zhang. "Hold Onto That Thought: Assessing KV Cache Compression on Reasoning." Efficient Reasoning Workshop, NeurIPS 2025. OpenReview. arXiv.

W79 Zhou, Yuhang, Jing Zhu, Shengyi Qian, Zhuokai Zhao, Xiyao Wang, Xiaoyu Liu, Ming Li, Paiheng Xu, Wei Ai, and Furong Huang. "DISCO Balances the Scales: Adaptive Domain- and Difficulty-Aware Reinforcement Learning on Imbalanced Data." Efficient Reasoning Workshop, NeurIPS 2025. OpenReview. arXiv.

W78 Ding, Mucong, Sean Michael McLeish, Kazem Meidani, Igor Melnyk, Nam H Nguyen, C. Bayan Bruss, and Furong Huang. "You Only Train Once: Efficient Tokenizer Selection for Arithmetic in Language Models." Tokenization Workshop (TokShop), ICML 2025.

W77 Yu, Kelin, Sheng Zhang, Harshit Soora, Furong Huang, Heng Huang, Pratap Tokekar, and Ruohan Gao. "GENFLOWRL: Generative Object-Centric Flow Matching for Reward Shaping in Visual Reinforcement Learning." ICRA 2025 Workshop on Foundation Models and Neuro-Symbolic AI for Robotics, spotlight, ICRA 2025.

W76 Liang, Yongyuan, Xiyao Wang, Yuanchen Ju, Jianwei Yang, and Furong Huang. "LEMON: A Unified and Scalable 3D Multimodal Model for Universal Spatial Understanding." The 4th Workshop on Computer Vision in the Wild, spotlight, CVPR 2025.

W75 Zheng, Ruijie, Jing Wang, Scott Reed, Johan Bjorck, Yu Fang, Fengyuan Hu, Joel Jang, Kaushil Kundalia, Zongyu Lin, Loïc Magne, Avnish Narayan, You Liang Tan, Guanzhi Wang, Qi Wang, Jiannan Xiang, Yinzhe Xu, Seonghyeon Ye, Jan Kautz, Furong Huang, Yuke Zhu, and Linxi Fan. "FLARE: Robot Learning with Implicit World Modeling." The 4th Workshop on Computer Vision in the Wild, spotlight, CVPR 2025.

W74 Zheng, Ruijie, Jing Wang, Scott Reed, Johan Bjorck, Yu Fang, Fengyuan Hu, Joel Jang, Kaushil Kundalia, Zongyu Lin, Loïc Magne, Avnish Narayan, You Liang Tan, Guanzhi Wang, Qi Wang, Jiannan Xiang, Yinzhe Xu, Seonghyeon Ye, Jan Kautz, Furong Huang, Yuke Zhu, and Linxi Fan. "FLARE: Robot Learning with Implicit World Modeling." Structured World Models for Robotic Manipulation (SWOMO), oral, RSS 2025.

W73 Chiang, Jeffrey Yang Fan, Seungjae Lee, Jia-Bin Huang, Furong Huang, and Yizheng Chen. "Why Are Web AI Agents More Vulnerable Than Standalone LLMs? A Security Analysis." Workshop BuildingTrust Workshop, ICLR 2025.

W72 Cai, Zikui, Shayan Shabihi, Bang An, Zora Che, Brian R. Bartoldson, Bhavya Kailkhura, Tom Goldstein, and Furong Huang. "AegisLLM: Scaling Agentic Systems for Self-Reflective Defense in LLM Security." Workshop BuildingTrust Workshop, ICLR 2025.

W71 Pathmanathan, Pankayaraj, Udari Madhushani Sehwag, Michael-Andrei Panaitescu-Liess, and Furong Huang. "AdvB-DGen: A Robust Framework for Generating Adaptive and Stealthy Backdoors in LLM Alignment Attacks." Workshop BuildingTrust Workshop, ICLR 2025.

W70 Wang, Xiyao, Zhengyuan Yang, Linjie Li, Hongjin Lu, Yuancheng Xu, Chung-Ching Lin, Kevin Lin, Furong Huang*, and Lijuan Wang*. "VisVM: Scaling Inference-Time Search with Vision Value Model for Improved Visual Comprehension." Scaling Self-Improving Foundation Models without Human Supervision (SSI-FM), ICLR 2025.

W69 Wang, Xiyao, Zhengyuan Yang, Linjie Li, Hongjin Lu, Yuancheng Xu, Chung-Ching Lin, Kevin Lin, Furong Huang*, and Lijuan Wang*. "VisVM: Scaling Inference-Time Search with Vision Value Model for Improved Visual Comprehension." Workshop on World Models, ICLR 2025.

W68 Wang, Xiyao, Zhengyuan Yang, Linjie Li, Hongjin Lu, Yuancheng Xu, Chung-Ching Lin, Kevin Lin, Furong Huang*, and Lijuan Wang*. "VisVM: Scaling Inference-Time Search with Vision Value Model for Improved Visual Comprehension." Reasoning and Planning for LLMs, ICLR 2025.

W67 Zheng, Ruijie, Yongyuan Liang, Shuaiyi Huang, Jianfeng Gao, Hal Daumé III, Andrey Kolobov, Furong Huang, and Jianwei Yang. "TraceVLA: Visual Trace Prompting Enhances Spatial-Temporal Awareness for Generalist Robotic Policies." Workshop on Generative Models for Robot Learning (GenBot), ICLR 2025.

W66 Pathmanathan, Pankayaraj, Udari Madhushani Sehwag, Michael-Andrei Panaitescu-Liess, and Furong Huang. "AdvBDGen: Adversarially Fortified Prompt-Specific Fuzzy Backdoor Generator Against LLM Alignment." The AAAI-25 Workshop on Artificial Intelligence for Cyber Security, AICS 2025.

W65 Panaitescu-Liess, Michael-Andrei, Zora Che, Bang An, Yuancheng Xu, Pankayaraj Pathmanathan, Souradip Chakraborty, Sicheng Zhu, Tom Goldstein, and Furong Huang. "Can Watermarking Large Language Models Prevent Copyrighted Text Generation and Hide Training Data?" **Best Paper Award**, Oral, The Third Workshop on New Frontiers in Adversarial Machine Learning, NeurIPS 2024.

W64 Panaitescu-Liess, Michael-Andrei, Pankayaraj Pathmanathan, Yigitcan Kaya, Zora Che, Bang An, Sicheng Zhu, Aakriti Agrawal, and Furong Huang. "PoisonedParrot: Subtle Data Poisoning Attacks to Elicit Copyright-Infringing Content from Large Language Models." Safe Generative AI Workshop, NeurIPS 2024.

W63 Agrawal, Aakriti, Mucong Ding, Zora Che, Chenghao Deng, Anirudh Satheesh, John Langford, and Furong Huang. "EnsemW2S: Can an Ensemble of LLMs be Leveraged to Obtain a Stronger LLM?" Safe Generative AI Workshop, NeurIPS 2024.

W62 Che, Zora, Stephen Casper, Anirudh Satheesh, Rohit Gandikota, Domenic Rosati, Stewart Slocum, Lev E McKinney,

Zichu Wu, Zikui Cai, Bilal Chughtai, Furong Huang, and Dylan Hadfield-Menell. "Model Manipulation Attacks Enable More Rigorous Evaluations of LLM Unlearning." Safe Generative AI Workshop, NeurIPS 2024.

W61 Pathmanathan, Pankayaraj, Udari Madhushani Sehwag, Michael-Andrei Panaitescu-Liess, and Furong Huang. "AdvB-DGen: Adversarially Fortified Prompt-Specific Fuzzy Backdoor Generator Against LLM Alignment." Safe Generative AI Workshop, NeurIPS 2024.

W60 Liang, Yongyuan, Tingqiang Xu, Kaizhe Hu, Guangqi Jiang, Furong Huang, and Huazhe Xu. "Make-An-Agent: A Generalizable Policy Network Generator with Behavior-Prompted Diffusion." Adaptive Foundation Models: Evolving AI for Personalized and Efficient Learning, oral, NeurIPS 2024.

W59 Liu, Xiaoyu, Jiaxin Yuan, Yuhang Zhou, Jingling Li, Furong Huang, and Wei Ai. "CSRec: Rethinking Sequential Recommendation from A Causal Perspective." Causal Representation Learning Workshop, NeurIPS 2024.

W58 Bornstein, Marco, Tahseen Rabbani, Brian Joseph Gravelle, and Furong Huang. "Shrinking the Size of Extreme Multi-Label Classification." Workshop on Machine Learning and Compression, NeurIPS 2024.

W57 Rabbani, Tahseen, Minghui Liu, Tony O'Halloran, Ananth Sankaralingam, Mary-Anne Hartley, and Furong Huang. "LSH-E Tells You What To Discard: An Adaptive Locality-Sensitive Strategy for KV Cache Compression." Workshop on Machine Learning and Compression, NeurIPS 2024.

W56 Chakraborty, Souradip, Jiahao Qiu, Hui Yuan, Alec Koppel, Furong Huang, Dinesh Manocha, Amrit Bedi, and Mengdi Wang. "MaxMin-RLHF: Towards Equitable Alignment of Large Language Models with Diverse Human Preferences." Oral, ICML 2024 Workshop on Models of Human Feedback for AI Alignment, ICML 2024.

W55 Pathmanathan, Pankayaraj, Souradip Chakraborty, Xiangyu Liu, Yongyuan Liang, and Furong Huang. "Is poisoning a real threat to LLM alignment? Maybe more so than you think." ICML 2024 Workshop on Models of Human Feedback for AI Alignment, ICML 2024.

W54 Panaitescu-Liess, Michael-Andrei, Zora Che, Bang An, Yuancheng Xu, Pankayaraj Pathmanathan, Souradip Chakraborty, Sicheng Zhu, Tom Goldstein, and Furong Huang. "Can Watermarking Large Language Models Prevent Copyrighted Text Generation and Hide Training Data?" ICML 2024 Next Generation of AI Safety Workshop, ICML 2024.

W53 An, Bang, Sicheng Zhu, Ruiyi Zhang, Michael-Andrei Panaitescu-Liess, Yuancheng Xu, and Furong Huang. "Automatic Pseudo-Harmful Prompt Generation for Evaluating False Refusals in Large Language Models." ICML 2024 Next Generation of AI Safety Workshop, ICML 2024.

W52 Ding, Mucong, Souradip Chakraborty, Vibhu Agrawal, Zora Che, Alec Koppel, Mengdi Wang, Amrit Bedi, and Furong Huang. "SAIL: Self-improving Efficient Online Alignment of Large Language Models." ICML 2024 Workshop on Theoretical Foundations of Foundation Models, ICML 2024. arXiv.

W51 Chakraborty, Souradip, Soumya Suvra Ghosal, Ming Yin, Dinesh Manocha, Mengdi Wang, Amrit Bedi, and Furong Huang. "Transfer Q-star: Principled Decoding for LLM Alignment." Adaptive Learning in Complex Environments TTIC Workshop 2024.

W50 Ding, Mucong, Tahseen Rabbani, Bang An, Aakriti Agrawal, Yuancheng Xu, Chenghao Deng, Sicheng Zhu, Abdirisak Mohamed, Yuxin Wen, Tom Goldstein, and Furong Huang. "WAVES: Benchmarking the Robustness of Image Watermarks." Workshop on Reliable and Responsible Foundation Models, ICLR 2024.

W49 Ding, Mucong, Tahseen Rabbani, Bang An, Aakriti Agrawal, Yuancheng Xu, Chenghao Deng, Sicheng Zhu, Abdirisak Mohamed, Yuxin Wen, Tom Goldstein, and Furong Huang. "WAVES: Benchmarking the Robustness of Image Watermarks." Workshop on Privacy Regulation and Protection in Machine Learning, ICLR 2024.

W48 Xu, Yuancheng, Jiarui Yao, Manli Shu, Yanchao Sun, Zichu Wu, Ning Yu, Tom Goldstein, and Furong Huang. "Shadowcast: Stealthy Data Poisoning Attacks Against Vision-Language Models." Workshop on Secure and Trustworthy Large Language Models, ICLR 2024.

W47 Xu, Yuancheng, Jiarui Yao, Manli Shu, Yanchao Sun, Zichu Wu, Ning Yu, Tom Goldstein, and Furong Huang. "Shadowcast: Stealthy Data Poisoning Attacks Against Vision-Language Models." Workshop on Navigating and Addressing Data Problems for Foundation Models, ICLR 2024.

W46 Bornstein, Marco, Amrit Bedi, Anit Kumar Sahu, Furqan Khan, and Furong Huang. "RealFM: A Realistic Mechanism to Incentivize Data Contribution and Device Participation." Workshop on Federated Learning in the Age of Foundation Models (FL@FM-NeurIPS'23), NeurIPS 2023.

W45 Wang, Xiyao, Ruijie Zheng, Yanchao Sun, Ruonan Jia, Wichayaporn Wongkamjan, Huazhe Xu, and Furong Huang. "COPlanner: Plan to Roll Out Conservatively but to Explore Optimistically for Model-Based RL." Workshop on Generalization in Planning (GenPlan '23), NeurIPS 2023.

W44 Nguyen, Khanh, Ruijie Zheng, Hal Daume III, Furong Huang, and Karthik Narasimhan. "Progressively Efficient Communication." Intrinsically Motivated Open-ended Learning (IMOL) Workshop, NeurIPS 2023.

W43 Zheng, Ruijie, Yongyuan Liang, Xiyao Wang, Shuang Ma, Hal Daume III, Huazhe Xu, John Langford, Praveen Palanisamy, Kalyan Basu, and Furong Huang. "PREMIER-TACO is a Few-Shot Policy Learner: Pretraining Multitask Representation via Temporal Action-Driven Contrastive Loss." 6th Robot Learning Workshop: Pretraining, Fine-Tuning, and Generalization with Large Scale Models, NeurIPS 2023.

W42 Zheng, Ruijie, Yongyuan Liang, Xiyao Wang, Shuang Ma, Hal Daume III, Huazhe Xu, John Langford, Praveen

Palanisamy, Kalyan Basu, and Furong Huang. "PREMIER-TACO is a Few-Shot Policy Learner: Pretraining Multitask Representation via Temporal Action-Driven Contrastive Loss." Workshop on Foundation Models for Decision Making, oral, NeurIPS 2023.

W41 Liu, Xiangyu, Chenghao Deng, Yanchao Sun, Yongyuan Liang, and Furong Huang. "Beyond Worst-case Attacks: Robust RL with Adaptive Defense via Non-dominated Policies." Workshop on Multi-Agent Security: Security as Key to AI Safety, spotlight, NeurIPS 2023.

W40 Zhu, Sicheng, Ruiyi Zhang, Bang An, Gang Wu, Joe Barrow, Zichao Wang, Furong Huang, Ani Nenkova, and Tong Sun. "AutoDAN: Automatic and Interpretable Adversarial Attacks on Large Language Models." Workshop on Socially Responsible Language Modelling Research (SoLaR), NeurIPS 2023.

W39 Agrawal, Aakriti, Rohith Aralikatti, Yanchao Sun, and Furong Huang. "Robustness to Multi-Modal Environment Uncertainty in MARL using Curriculum Learning." Workshop Multi-Agent Security: Security as Key to AI Safety, spotlight, NeurIPS 2023.

W38 Xu, Yuancheng, Chenghao Deng, Yanchao Sun, Ruijie Zheng, Xiyao Wang, Jieyu Zhao, and Furong Huang. "Equal Long-term Benefit Rate: Adapting Static Fairness Notions to Sequential Decision Making." AdvML-Frontiers workshop, ICML 2023.

W37 Liang, Yongyuan, Yanchao Sun, Ruijie Zheng, Xiangyu Liu, Tuomas Sandholm, Furong Huang, and Stephen McAleer. "Adapting Robust Reinforcement Learning to Handle Temporally-Coupled Perturbations." AdvML-Frontiers workshop, ICML 2023.

W36 An, Bang, Sicheng Zhu, Michael-Andrei Panaitescu-Liess, Chaithanya Kumar Mummadi, and Furong Huang. "Mental Calibration: Discovering and Adjusting for Latent Factors Improves Zero-Shot Inference of CLIP." Workshop on Efficient Systems for Foundation Models (ES-FoMO), ICML 2023.

W35 Chakraborty, Souradip, Amrit Singh Bedi, Alec Koppel, Furong Huang, and Mengdi Wang. "Principal-Driven Reward Design and Agent Policy Alignment via Bilevel-RL." Interactive Learning with Implicit Human Feedback Workshop (ILHF), ICML 2023.

W34 Liu, Xiaoyu, Jiaxin Yuan, Bang An, Yuancheng Xu, Yifan Yang, and Furong Huang. "C-Disentanglement: Discovering Causally-Independent Generative Factors under an Inductive Bias of Confounder." The Second Workshop on Spurious Correlations, Invariance and Stability (SCIS), ICML 2023.

W33 Liu, Xiaoyu, Jiaxin Yuan, Bang An, Yuancheng Xu, Yifan Yang, and Furong Huang. "C-Disentanglement: Discovering Causally-Independent Generative Factors under an Inductive Bias of Confounder." Workshop on Structured Probabilistic Inference & Generative Modeling (SPIGM), ICML 2023.

W32 Ding, Peijian, Davit Soselia, Thomas Armstrong, Jiahao Su, and Furong Huang. "Reviving Shift Equivariance in Vision Transformers." The Second Workshop on Spurious Correlations, Invariance and Stability (SCIS), ICML 2023.

W31 Zhu, Sicheng, Bang An, Furong Huang, and Sanghyun Hong. "Learning Unforeseen Robustness from Out-of-distribution Data Using Equivariant Domain Translator." Workshop on Trustworthy ML, ICLR 2023.

W30 Rabbani, Tahseen, Marco Bornstein, and Furong Huang. "PGHash: Large-Scale Distributed Learning via Private On-Device Locally Sensitive Hashing." Sparsity in Neural Networks 2023 workshop, ICLR 2023.

W29 Sun, Yanchao, Ruijie Zheng, Parisa Hassanzadeh, Yongyuan Liang, Soheil Feizi, Sumitra Ganesh, and Furong Huang. "Certifiably Robust Policy Learning against Adversarial Multi-Agent Communication." Rebellion and Disobedience in AI (RaD-AI) Workshop, AAMAS 2023.

W28 Chakraborty, Souradip, Amrit Singh Bedi, Alec Koppel, Furong Huang, and Dinesh Manocha. "Stein Information Directed Sampling for Efficient Exploration in Model-Based Reinforcement Learning." Oral, AAAI Reinforcement Learning Ready for Production Workshop, 2023.

W27 Bornstein, Marco, Tahseen Rabbani, Evan Wang, Amrit Singh Bedi, and Furong Huang. "SWIFT: Rapid Decentralized Federated Learning via Wait-Free Model Communication." FL-NeurIPS Workshop, oral (11%), NeurIPS 2022.

W26 Zheng, Ruije, Xiyao Wang, Huazhe Xu, and Furong Huang. "Is Model Ensemble Necessary? Model-based RL via a Single Model with Lipschitz Regularized Value Function." Deep Reinforcement Learning Workshop, oral, NeurIPS 2022.

W25 Ding, Mucong, Xiaoyu Liu, Tahseen Rabbani, and Furong Huang. "Faster Hyperparameter Search on Graphs via Calibrated Dataset Condensation." GLFrontiers Workshop, NeurIPS 2022.

W24 Ding, Mucong, Tahseen Rabbani, Bang An, Evan Wang, and Furong Huang. "Sketch-GNN: Scalable Graph Neural Networks with Sublinear Training Complexity." GLFrontiers Workshop, NeurIPS 2022.

W23 Sun, Yanchao, Shuang Ma, Ratnesh Madaan, Rogerio Bonatti, Furong Huang, and Ashish Kapoor. "SMART: Self-supervised Multi-task pretrAining with contRol Transformers." Foundation Models for Decision Making Workshop, NeurIPS 2022.

W22 Chakraborty, Souradip, Amrit Singh Bedi, Alec Koppel, Furong Huang, Pratap Tokekar, and Dinesh Manocha. "Posterior Coreset Construction with Kernelized Stein Discrepancy for Model-Based Reinforcement Learning." Workshop on Score-Based Methods, NeurIPS 2022.

W21 Liu, Xiangyu, Souradip Chakraborty, and Furong Huang. "Controllable Attack and Improved Adversarial Training in Multi-Agent Reinforcement Learning." Workshop on Trustworthy and Socially Responsible Machine Learning (TSRML),

Outstanding Paper Award, Oral, NeurIPS 2022.

W20 Xu, Paiheng, Yuhang Zhou, Bang An, Wei Ai, and Furong Huang. "GFairHint: Improving Individual Fairness for Graph Neural Networks via Fairness Hint." workshop on Trustworthy and Socially Responsible Machine Learning (TSRML), NeurIPS 2022.

W19 Borgnia, Eitan, Jonas Geiping, Valeriia Cherepanova, Liam H. Fowl, Arjun Gupta, Amin Ghiasi, Furong Huang, Micah Goldblum, and Tom Goldstein. "DP-InstaHide: Data Augmentations Provably Enhance Guarantees Against Dataset Manipulations." ML Safety workshop, NeurIPS 2022.

W18 Sun, Yanchao, Ruijie Zheng, Parisa Hassanzadeh, Yongyuan Liang, Soheil Feizi, Sumitra Ganesh, and Furong Huang. "Certifiably Robust Multi-Agent Reinforcement Learning against Adversarial Communication." Workshop on Responsible Decision Making in Dynamic Environments, ICML 2022.

W17 Liang, Yongyuan, Yanchao Sun, Ruijie Zheng, and Furong Huang. "Efficient Adversarial Training without Attacking: Worst-Case-Aware Robust Reinforcement Learning." Workshop on Responsible Decision Making in Dynamic Environments, ICML 2022.

W16 Xu, Yuancheng, Yanchao Sun, and Furong Huang. "Everyone Matters: Customizing the Dynamics of Decision Boundary for Adversarial Robustness." Workshop on Continuous Time Methods for Machine Learning, ICML 2022.

W15 Wang, Xiyao, Wichayaporn Wongkamjan, and Furong Huang. "Live in the Moment: Learning Dynamics Model Adapted to Evolving Policy." Workshop on Decision Awareness in Reinforcement Learning (DARL), Spotlight, ICML 2022.

W14 An, Bang, Zora Che, Mucong Ding, and Furong Huang. "Transfer Fairness under Distribution Shifts." Workshop on Socially Responsible Machine Learning, ICLR 2022.

W13 Rabbani, Tahseen, Brandon Feng, Yifan Yang, Arjun Rajkumar, Amitabh Varshney, and Furong Huang. "Comfetch: Federated Learning of Large Networks on Memory-Constrained Clients via Sketching." Workshop on Trustable, Verifiable and Auditable Federated Learning, AAAI 2022.

W12 Sun, Yanchao, Ruijie Zheng, Yongyuan Liang, and Furong Huang. "Who Is the Strongest Enemy? Towards Optimal and Efficient Evasion Attacks in Deep RL." Workshop on Safe and Robust Control of Uncertain Systems (SafeRL), **Best Paper Award**, Oral, NeurIPS 2021.

W11 Liang, Yongyuan, Yanchao Sun, Ruijie Zheng, and Furong Huang. "Efficiently Improving the Robustness of RL Agents against Strongest Adversaries." Workshop on Safe and Robust Control of Uncertain Systems (SafeRL), Oral, NeurIPS 2021.

W10 Ding, Mucong, Kezhi Kong, Jiuhai Chen, John Kirchenbauer, Micah Goldblum, David Wipf, Furong Huang, and Tom Goldstein. "A Closer Look at Distribution Shifts and Out-of-Distribution Generalization on Graphs." Workshop on Distribution Shifts: Connecting Methods and Applications (DistShift), Spotlight, NeurIPS 2021.

W9 Yousefzadeh, Roozbeh, and Furong Huang. "Interpretable Insights about Medical Image Datasets: Using Wavelets and Spectral Methods." Workshop on Human Interpretability in Machine Learning (WHI), ICML 2020.

W8 Huang, Furong, Yu-Xiang Wang, and Xuchen You. "A Spectral Method for Off-Policy Evaluation in Contextual Bandits under Distribution Shift." 13th Annual Machine Learning Symposium, 2019.

W7 Su, Jiahao, Milan Cvitkovic, and Furong Huang. "Sampling-Free Learning of Bayesian Quantized Neural Networks." 13th Annual Machine Learning Symposium, 2019.

W6 Su, Jiahao, Jingling Li, Bobby Bhattacharjee, and Furong Huang. "Tensorial Neural Networks: Generalization of Neural Networks and Application to Model Compression." 13th Annual Machine Learning Symposium, 2019. arXiv.

W5 Li, Jingling, Yanchao Sun, Jiahao Su, Taiji Suzuki, and Furong Huang. "Understanding of Generalization in Deep Learning via Tensor Methods." Workshop on Generalization, ICML 2019.

W4 Su, Jiahao, Jingling Li, Bobby Bhattacharjee, and Furong Huang. "Tensorial Neural Networks: Generalization of Neural Networks and Application to Model Compression." Women in Machine Learning Workshop, NeurIPS 2018. arXiv.

W3 Xu, Zheng, Furong Huang, Louiqa Raschid, and Tom Goldstein. "Non-negative Factorization of the Occurrence Tensor from Financial Contracts." Tensor-Learn Workshop, NIPS 2016.

W2 Huang, Furong, Niranjan U.N., Ioakeim Perros, Robert Chen, Jimeng Sun, and Animashree Anandkumar. "Scalable Latent Tree Model and its Application to Health Analytics." Machine Learning in Healthcare Workshop, NIPS 2015.

W1 Huang, Furong, Sergiy Matusevych, Anima Anandkumar, Nikos Karampatziakis, and Paul Mineiro. "Distributed Latent Dirichlet Allocation via Tensor Factorization." Optimization for Machine Learning Workshop, NIPS 2014.

Datasets and Benchmarks

20. **MomaGraph-Scenes and MomaGraph-Bench.** Large-scale dataset of richly annotated, task-driven spatial-functional scene graphs for household environments, paired with a benchmark spanning six embodied-reasoning capabilities. Project page; dataset; benchmark.

19. **TraceForge-123K and TraceGen Benchmark.** Cross-embodiment corpus of 123K videos and 1.8M observation–trace–language triplets, with held-out evaluation across five human- and robot-video environments. Project page; dataset collection; benchmark.

18. μ_0 **Evaluation Dataset.** Released evaluation episodes and normalization statistics for assessing transferable 3D interaction-trace prediction across human and robot embodiments. Project page; evaluation dataset and model artifacts.
17. **SoundnessBench.** Benchmark of 1,099 machine-learning research proposals for evaluating whether AI research agents can assess proposal-stage methodological soundness. Project page; dataset.
16. **TSRBench.** Multi-task, multimodal time-series reasoning benchmark for generalist models. Benchmark and project page.
15. **ROVER.** Benchmark for reciprocal cross-modal reasoning across image, video, audio, and 3D understanding and generation. Project page; code.
14. **PropensityBench.** Agentic evaluation suite for latent safety risks in large language models. Code and benchmark.
13. **Zebra-CoT.** Dataset for interleaved vision-language reasoning. Dataset.
12. **TrustGen.** Dynamic benchmarking platform for evaluating trustworthiness across text-to-image, language, and vision-language foundation models. Platform documentation; paper.
11. **Sequential Embodied Question Answering (EQA).** Evaluation setting for testing persistent visual-semantic memory across multiple questions and continuous embodied operation. Paper and benchmark description.
10. **MORSE-500.** Programmatically controllable video benchmark with 500 scripted clips for stress-testing abstract, physical, planning, spatial, and temporal multimodal reasoning. Project page; paper.
9. **Erasing the Invisible.** NeurIPS competition benchmark, dataset, and evaluation toolkit for stress-testing image watermarks under black-box and beige-box attacks. Project page; competition.
8. **Fictitious Facial Identity Dataset.** Benchmark for evaluating vision-language-model unlearning of fictitious identities. Paper and dataset description.
7. **Easy2Hard-Bench.** Standardized difficulty labels and evaluation tasks for profiling language-model performance and easy-to-hard generalization. Project page.
6. **Mementos.** Benchmark and dataset for multimodal language-model reasoning over coherent image sequences. Project page; dataset.
5. **HallusionBench.** Diagnostic benchmark for entangled language hallucination and visual illusion in large vision-language models. Benchmark and code.
4. **AutoHallusion.** Automatically generated hallucination benchmarks for diagnosing vision-language models across diverse visual reasoning failures. Benchmark and code.
3. **WAVES.** Standardized benchmark and stress-testing toolkit for evaluating image-watermark robustness across detection and identification tasks. Project page.
2. **PHTest.** Evaluation dataset of pseudo-harmful prompts for measuring false refusals and the safety–helpfulness tradeoff in language models. Dataset; code.
1. **Easy-to-Hard Generalization Datasets.** Controlled image-classification datasets for studying generalization from easy training examples to harder test examples. Paper and datasets.

Awards and Honors

28. NVIDIA Academic Grant, 2026.
27. Best Paper Award at the New Frontiers in Adversarial Machine Learning (AdvML Frontier) Workshop, NeurIPS 2024.
26. National Science Foundation (NSF) National Artificial Intelligence Research Resource (NAIRR) Pilot project NAIRR240045 “Guardians of Integrity in AI: Establishing Trust, Originality, and Ethical Standards (GATES),” 2024.
25. Adobe Research Gift Grant, 2024.
24. Microsoft Accelerate Foundation Models Research award, 2023.
23. CloudBank Computing Award for Research on Fair Machine Learning in the Long Run, 2023.
22. Outstanding Paper Award at the Trustworthy and Socially Responsible Machine Learning Workshop, NeurIPS 2022.
21. Winner of MIT Technology Review Innovators Under 35 (TR35) Asia Pacific, 2022.
20. Finalist of AI in Research - AI researcher of the year, 2022 Women in AI Awards North America, 2022.
19. Special Jury Recognition of Women in AI Awards North America, 2022.
18. JP Morgan Faculty Research Award, “Repelling Security Vulnerabilities in AI-augmented Financial Decision-Making Systems,” 2022.
17. Best Paper Award at Safe and Robust Control of Uncertain Systems, NeurIPS 2021.
16. IEEE Senior Member, Jan. 2021.
15. JP Morgan Faculty Research Award, “Robust, Private and Fair ML for Financial Models,” 2020.
14. JP Morgan Faculty Research Award, “Methods to Identify Communities and Trading Behavior over Financial Data Streams,” 2019.

13. Capital One Faculty Research Award, “Unsupervised Learning via Probabilistic Models,” 2018.
12. Adobe Faculty Research Award, “Understanding User Features with Text, Social and Behavior Data through Deep Tensor Neural Network Learning,” 2017.
11. Travel Grant Award: Heidelberg Laureate Forum, 2017.
10. Travel Grant Award: CRA-W Early Career Mentoring Workshop, 2016.
9. MLconf Industry Impact Student Research winner sponsored by Google, San Francisco, 2015.
8. Travel Grant Award: NIPS, 2015.
7. Travel Grant Award: Junior Scientist Workshop on Machine Learning and Computer Vision, 2015.
6. Electrical Engineering and Computer Science Department Graduate Fellowship, University of California Irvine, 2010-2014.
5. Travel Grant Award: WiML, 2014.
4. Travel Grant Award: CRA-W Grad. Cohort Workshop, 2012.
3. Travel Grant Award: WiML, 2012.
2. Department Fellowship, 2010.
1. Best Undergraduate Thesis Award, Zhejiang University, 2010.

Grants

Total: $13,278,032, My Share: $9,229,399.

20. Feizi, S. (Lead Investigator), Huang, F. (Co-Investigator), Wilson, A. (Co-Investigator), “Consistency, Robustness and Generalization Guarantees for Understanding Language Models,” Sponsored by DARPA-Defense Advanced Research Projects Agency, $1,800,000. (July 1, 2025 - June 30, 2028).
19. Huang, F. (Lead Investigator), Goldstein, T. (Lead Investigator), Li, B. (Co-Investigator), Martin-Martin, R. (Co-Investigator), Zhu, Y. (Co-Investigator), “RAMSES: A Domain-Agnostic System with Robust Features and Symbolic Reasoning,” Sponsored by DARPA-Defense Advanced Research Projects Agency, $2,800,000. (July 1, 2024 - June 30, 2027).
18. Huang, F. (Single Lead Investigator), “Securing Retrieval-Augmented Generation (RAG) Systems,” Sponsored by Peraton Inc., $100,037. (01/01/2025 - 12/31/2025).
17. Huang, F. (Single Lead Investigator), “Encoded Reasoning in Agentic Systems, FAKE-LLM: Knowledgeable Exploitation, and Reasoning-Trace Augmented Training,” Sponsored by Good Ventures (recommended by Open Philanthropy), $1,118,700. (October 1, 2025–September 30, 2026).
16. Huang, F. (Single Lead Investigator), “Robust and Efficient Machine Learning Project 2,” Sponsored by Booz Allen and Hamilton, Inc. (Prime: Air Force Material Command), $236,178. (Mar 15, 2024 - Mar 14, 2025).
15. Huang, F. (Single Lead Investigator), “E-VERIFY: Robust and Efficient Ultra-large Sized Graph Learning - Project 13,” Sponsored by Booz Allen and Hamilton, Inc. (Prime: Air Force Material Command), $150,000. (Feb 13, 2023 - Feb 12, 2024).
14. Huang, F. (Single Lead Investigator), “Autonomous Decision-Making Systems under (Adversarial) Distribution Shifts: Learning to Adapt, Generalize and Robustify,” Sponsored by the Air Force Office of Scientific Research (AFOSR), $517,881. (December 15, 2022–December 14, 2025).
13. Huang, F. (Single Lead Investigator), “Repelling Security Vulnerabilities in AI-Augmented Financial Decision-Making Systems,” Sponsored by J.P. Morgan, $80,000. (Fall 2022–Fall 2023).
12. Huang, F. (Single Lead Investigator), “Trustworthy Real-time Communicative Multi-Agent Reinforcement Learning for Decision-making in Unstable & Adversarial Systems”, Sponsored by Office of Naval Research, $863,877. (June 1, 2022 - May 30, 2025).
11. Huang, F. (Lead Investigator), Wu, M. (Co-Investigator), Dachman-Soled D. (Co-Investigator), “FAI: Toward Fair Decision Making and Resource Allocation with Application to AI-Assisted Graduate Admission and Degree Completion”, Sponsored by National Science Foundation and Amazon, $1,000,000. (February 15, 2022 - January 31, 2025).
10. Huang, F. (Single Lead Investigator), “E-VERIFY: Task 11: Efficient and Expressive State-of-the-art Deep Learning Model Design through Tensor Methods,” Sponsored by Booz Allen and Hamilton, Inc. (Prime: Air Force Material Command), $180,000. (June 1, 2022 - May 30, 2023).
9. Huang, F. (Single Lead Investigator), “CRII: RI: Principled Methods for Learning and Understanding of Neural Networks,” Sponsored by National Science Foundation, $175,000. (May 1, 2019 - April 30, 2022).
8. Huang, F. (Single Lead Investigator), “E-VERIFY: Task 10: TNN Based Machine Learning 2 (FY21),” Sponsored by Booz Allen and Hamilton, Inc. (Prime: Air Force Material Command), $ 180,000. (August 23, 2021 - August 22, 2022).
7. Goldstein, T. (Lead Investigator), Huang, F. (Co-Investigator), “Robust, Private and Fair ML for Financial Models,” Sponsored by J. P. Morgan, $150,000 (my share: $75,000). (May 1, 2020 - August 31, 2021).

6. Raschid, L. (Lead Investigator), Huang, F. (Co-Investigator), Rossi, A. (Co-Investigator), "Methods to Identify Communities and Trading Behavior over Financial Data Streams," Sponsored by J. P. Morgan, $150,000 (my share: $75,000). (March 4, 2019 - March 3, 2021).
5. Huang, F. (Single Lead Investigator), "E-VERIFY: Task 6: High-Performance Machine Learning with Tensorial Neural Networks on Heterogeneous Systems," Sponsored by Booz Allen and Hamilton, Inc. (Prime: Air Force Material Command), $180,000. (January 1, 2020 - January 31, 2021).
4. Goldstein, T. (Lead Investigator), Katz, J. (Co-Investigator), Huang, F. (Co-Investigator), Shrivastava, A. (Co-Investigator), Jacobs, D. (Co-Investigator), Dickerson, J. (Co-Investigator), "Repelling Evasion and Poisoning Attacks: A Principled Way Forward," Sponsored by DOD-DARPA-Defense Advanced Research Projects Agency, $3,226,359 (my share: $537,726). (December 12, 2019 - November 30, 2023).
3. Raschid, L. (Lead Investigator), Huang, F. (Co-Investigator), "Customizing Machine Learning Approaches to Interpret Patterns in Financial Data Streams," Sponsored by AI in Business and Society Seed Grant, $20,000 (my share: $10,000). (2018).
2. Huang, F. (Lead Investigator), "Unsupervised Learning via Probabilistic Models," Sponsored by Capital One, $300,000. (2018–present).
1. Huang, F. (Single Lead Investigator), "Understanding User Features with Text, Social and Behavior Data through Deep Tensor Neural Network Learning," Sponsored by Adobe, $50,000. (2017).

Academic Talks

94. Tutorial speaker, "Foundations and Frontiers of Watermarking: Algorithms, Multimodal Extensions, Benchmarks, and Authenticity Frameworks," ACM Multimedia 2026, virtual presentation, Nov. 2026.
93. Distinguished Seminar, Department of Mechanical Engineering, Virginia Tech, virtual, Oct. 16, 2026. Talk title forthcoming.
92. Invited speaker, University of Maryland Computer Science Fall Seminar, College Park, MD, Fall 2026.
91. Featured speaker at the Agentic AI Summit 2026, "Reasoning as Control: Toward Self-Improving Agentic Systems Across Thinking, Actions, and Workflow," University of California, Berkeley, Berkeley, CA, Aug. 2026.
90. Panelist, "Vision, Multimodal AI & Agents," KAIST@ICML 2026, Seoul, South Korea, Jul. 2026.
89. Invited speaker at the ITGenNexus Workshop, IEEE International Symposium on Information Theory (ISIT), "Reasoning as Control: Adaptive Test-Time Compute for Planning Agents," Guangzhou, China, Jul. 2026.
88. Panelist, "From Information to Intelligence: What Can These Communities Learn from Each Other?," ITGenNexus Workshop at IEEE ISIT, Guangzhou, China, Jul. 2026.
87. Lightning talk at the Workshop on Reasoning and Planning, hosted by Apple, "From More Thinking to Better Thinking: Adaptive Test-Time Control for Reasoning and Planning Agents," Jun. 2026.
86. Invited speaker at DatologyAI's Summer of Data Seminar, "A Few Early Steps Away: Building Self-Improving Foundation Models—Auditors, Actuators, and Amplifiers for Trustworthy AI," Jun. 2026.
85. Invited speaker at Scale AI's ML Research Seminar, "A Few Early Steps Away: Building Self-Improving Foundation Models—Auditors, Actuators, and Amplifiers for Trustworthy AI," Jun. 2026.
84. Invited speaker at the ICLR Workshop on Algorithmic Fairness Across Alignment Procedures and Agentic Systems (AFAA), "Rethinking Test-Time Thinking: From Token-Level Rewards to Robust Generative Agents," Apr. 2026.
83. Panelist at the ICLR Workshop on Algorithmic Fairness Across Alignment Procedures and Agentic Systems (AFAA), Apr. 2026.
82. Colloquium at Johns Hopkins University, "Rethinking Test-Time Compute," 2026.
81. Keynote speaker at RL Workshop at IISc, "From Perception to Action: World Model Learning for Generalist Agents", Jan., 2026.
80. Keynote speaker at NeurIPS Workshop on Generative AI in Finance, "LLMs Are Not Accountants: Data Without Disclosure via Rule-Grounded World Models for Financial Ledgers", NeurIPS, Dec., 2025.
79. Keynote speaker at the 4th workshop on Computer Vision in the Wild, "From Perception to Action: World Model Learning for Generalist Agents", CVPR, Jun., 2025.
78. Keynote speaker at the 2nd Texas Colloquium on Distributed Learning (TL;DR), "Rethinking Test-Time Thinking: From Token-Level Rewards to Robust Generative Agents", May., 2025.
77. Keynote speaker at the GenAI Watermarking Workshop, "Erasing the Invisible: Riding the WAVES of Watermark Robustness", at ICLR, Apr., 2025.
76. Keynote speaker at the World Models Workshop, "Foundation Model for Sequential Decision-Making: A World-Model Learning Perspective", at ICLR, Apr., 2025.
75. Spotlight speaker at the Alignment Workshop, "Aligning AI on the Fly," Singapore, Apr. 2025.
74. Invited Talk at the 2024 IEEE International Conference on Image Processing (ICIP) Workshop SPVis '24 – Security

and Privacy of Machine Learning-Based Vision Processing in Autonomous Systems, "Crafting and Cracking AI in the Shadows of Language- Poison Data and Jailbreak Prompts for LLMs," Oct., 2024.

73. Keynote speaker at the New York Academy of Sciences, 15th Annual Machine Learning Symposium, "Towards Generative AI Security: An Interplay of Stress-Testing and Alignment," Oct., 2024.
72. Invited talk at UT Austin AI Institute Seminar (IFML), "Foundation Models for Sequential Decision-Making," Oct., 2024.
71. Invited talk at Vanderbilt University ML Seminar, "Integrity in AI: Multi-Modality Approaches to Combat Misinformation for Content Authenticity," Sep., 2024.
70. Invited talk at Capital One, "Foundation Models for Sequential Decision-Making," Sep., 2024.
69. Invited talk at MSRNYC Seminar, "Foundation Models for Sequential Decision-Making," Aug., 2024.
68. Keynote speaker at the 1st CVPR Workshop on Dataset Distillation (DDCV) at CVPR, "Advancing AI with Data-Centric Strategies—Boosting Efficiency, Generalization, and Trust," Jun., 2024.
67. Invited talk at AI Community of Practice Seminar at the U.S. Securities and Exchange Commission (SEC), "Integrity in AI: Multi-Modality Approaches to Combat Misinformation for Content Authenticity," May, 2024.
66. Invited talk at JP Morgan, "Invisible Foes: Crafting and Cracking AI in the Shadows of Language," May, 2024.
65. Invited talk at Maryland Robotics Center (MRC) Symposium, "Foundation Models for Sequential Decision-Making," May, 2024.
64. Invited talk at Qualcomm AI Security Lecture Series, "Invisible Foes: Crafting and Cracking AI in the Shadows of Language — Poison Finetuning Data and Jailbreak Prompts for LLMs," Mar., 2024.
63. Seminar talk at Values-Centered Artificial Intelligence (VCAI) seminar series, "Algorithmic Fairness in an Ever-Changing World," College Park, MD, Feb., 2024.
62. Lightning talk at NSF-Amazon FAI PI meeting, "Algorithmic Fairness in an Ever-Changing World," Amazon HQ2, Arlington, VA, Jan., 2024.
61. Department Colloquium at University of Maryland, College Park, "Trustworthy Machine Learning in an Ever-Changing World," Sep., 2023.
60. Panelist at Interactive Learning with Implicit Human Feedback workshop, ICML, Jul., 2023.
59. Keynote speaker at ROADS to Mega-AI Models Workshop, MLSys, "Efficient Machine Learning at the Edge," Jun., 2023.
58. Invited talk at the 3rd Workshop of Adversarial Machine Learning on Computer Vision: Art of Robustness, "Robust Reinforcement Learning in an Ever-Changing World," CVPR, Jun., 2023.
57. Invited talk at Reincarnating RL workshop, "Adaptable Reinforcement Learning in An Ever-Changing World," ICLR, May., 2023.
56. Panelist at Reincarnating RL workshop, ICLR, May., 2023.
55. Spotlight talk at NSF-IEEE workshop: Toward Explainable, reliable, and sustainable machine learning in signal & data science, "Trustworthy Machine Learning in Complex Environments," Mar., 2023.
54. Invited talk at 57th Annual Conference on Information Science and Systems, CISS, "Efficient Machine Learning at the Edge in Parallel," Mar., 2023.
53. Invited talk at the 2023 Information Theory and Applications Workshop (ITA), "Trustworthy Machine Learning in Complex Environments," Feb. 2023.
52. Invited talk at UTSA Matrix AI Seminar, "Trustworthy Machine Learning in Complex Environments," Jan., 2023.
51. Invited talk at 2022 MSR Asia Theory Workshop on Data-driven Optimization, "Efficient Machine Learning at the Edge in Parallel," Nov., 2022.
50. Invited talk at UCSD, "Trustworthy Machine Learning in Complex Environments," Oct., 2022.
49. Invited talk at Caltech, "Trustworthy Machine Learning in Complex Environments," Oct., 2022.
48. Invited talk at UCLA, "Trustworthy Machine Learning in Complex Environments," Oct., 2022.
47. Invited talk at USC, "Trustworthy Machine Learning in Complex Environments," Oct., 2022.
46. Invited talk at UCI, "Trustworthy Machine Learning in Complex Environments," Oct., 2022.
45. Invited talk at SIAM Conference on Mathematics of Data Science (MDS22), "Data-driven Design of Transformers for Improved Generalization or Efficiency," Oct., 2022.
44. Contributed talk at ICML workshop Decision Awareness in Reinforcement Learning, "Live in the Moment: Learning Dynamics Model Adapted to Evolving Policy," Jul., 2022.
43. Invited talk at the Applied Statistics Symposium at the ICSA (International Chinese Statistical Association) International Conference, "Spectral Methods in Deep Learning," Sep., 2021.
42. Invited talk at Code Girls Club at Montgomery Blair High School, "Learning in High-dimensional Data," Jun., 2021.
41. Invited talk at Institute for Pure & Applied Mathematics (IPAM), Efficient Tensor Representations for Learning and Computational Complexity Workshop, " Understanding, Interpreting and Design Neural Network Models Through Tensor

Representation," May, 2021.

40. Invited talk at the annual East Coast Optimization Meeting (ECOM), "Escaping from saddle points using asynchronous coordinate gradient descent," Apr., 2021.
39. Invited talk at the SIAM CSE21 minisymposium "High-performance tensor computations in scientific computing and machine learning," "Tensors in Modern Machine Learning," Mar., 2021.
38. Invited talk at ARO Sponsored Workshop on Assured Autonomy, "Poisoning Attacks in Reinforcement Learning," Jun., 2020.
37. Invited talk at High Performance Computing and Data Analytics Workshop, "Efficient Machine Learning via Tensorial Neural Networks," Sep., 2019.
36. IEEE CIS/SPS lecture, "Understanding of Generalization in Deep Learning via Tensor Methods," May 1, 2019.
35. Invited talk at Virginia Tech AI-AI Workshop, AI Workshop, "Tensorial Neural Networks: Generalization of Neural Networks and Application to Model Compression," May, 2019.
34. Invited talk at ICERM Workshop, Theory and Practice in Machine Learning and Computer Vision, "Tensorial Neural Networks: Generalization of Neural Networks and Application to Model Compression," Feb., 2019.
33. Seminar talk at University of Montreal, "Theory in Deep Learning," Dec., 2018.
32. Invited talk at UMD-NCI Data Science Workshop, "Discovery of Latent Factors in High-dimensional Data via Spectral Methods," Nov., 2018.
31. Invited talk at the 55th Annual Allerton Conference on Communication, Control, and Computing, session on Large-Scale Optimization, "Principled Methods for Learning and Understanding of Neural Networks," Monticello, Illinois, Oct., 2018.
30. Invited talk at Workshop on Quantum Machine Learning, "Discovery of Latent Factors in High-dimensional Data via Spectral Methods," Sep., 2018.
29. Seminar talk at Virginia Tech, "Discovery of Latent Factors in High-dimensional Data via Spectral Methods," Oct., 2018.
28. Talk at Deep Learning RIT, "Principled Methods for Learning and Understanding of Neural Networks," University of Maryland, Oct., 2018.
27. Seminar talk at Laboratory for Telecommunication Science (LTS), "Guaranteed Learning of Latent Variable Models through Tensor Methods," Jul., 2018.
26. Invited tutorial at ACM Sigmetrics, Irvine, California, Jun., 2018.
25. Seminar talk at Laboratory for Telecommunication Science (LTS), "Fast Detection of Overlapping Communities via Online Tensor Methods on GPUs," Nov., 2017.
24. Undergraduate Honors Guest Lecture, "Matrices and Tensors in Machine Learning," Nov., 2017.
23. Seminar talk at the Department of Computer Science, "Guaranteed Learning of Latent Variable Models through Tensor Methods," Virginia Tech, Nov., 2017.
22. Department Colloquium at Computer Science Department, "Guaranteed Learning of Latent Variable Models through Tensor Methods," University of Maryland College Park, Oct., 2017.
21. Invited talk at the Matrix and Tensor Factorization Methods in Computer Vision, ICCV conference, Venice, Italy, Oct., 2017.
20. Invited talk at the 55th Annual Allerton Conference on Communication, Control, and Computing, session on Large-Scale Optimization, "Nonconvex Optimization in Tensor Decomposition," Monticello, Illinois, Oct., 2017.
19. Seminar talk at New York University, "Guaranteed Learning of Latent Variable Models through Tensor Methods," New York City, Apr., 2017.
18. Spotlight talk at 11th Annual Machine Learning Symposium, "Convolutional Dictionary Learning through Tensor Factorization," New York City, Mar., 2017.
17. Welcome and opening remark at the live streaming event for "Women in Data Science" conference at Stanford University, located in New York City, Feb., 2017.
16. Tea talk at Microsoft Research New York City, "Guaranteed Learning of Latent Variable Models through Tensor Methods," Nov., 2016.
15. Invited talk at Machine Learning conference, "Guaranteed Learning of Latent Variable Models through Tensor Methods," New York City, Apr., 2016.
14. Invited talk at University of Maryland, College Park, "Guaranteed Learning of Latent Variable Models through Tensor Methods," Mar., 2016.
13. Invited talk at IBM T. J. Watson Research Center in Yorktown Heights, "Guaranteed Learning of Latent Variable Models through Tensor Methods," New York, Feb., 2016.
12. Invited talk at Microsoft Research New York City, "Guaranteed Learning of Latent Variable Models through Tensor Methods," Feb., 2016.
11. Invited talk at Simons Foundation New York City, "Guaranteed Learning of Latent Variable Models through Tensor

Methods," Feb., 2016.

10. Invited talk at Boston University, "Guaranteed Learning of Latent Variable Models through Tensor Methods," Mar., 2016.
9. Invited talk at Tufts University, "Guaranteed Learning of Latent Variable Models through Tensor Methods," Mar., 2016.
8. Invited talk at University of California Davis, "Guaranteed Learning of Latent Variable Models through Tensor Methods," Mar., 2016.
7. Invited talk at Pennsylvania State University, "Guaranteed Learning of Latent Variable Models through Tensor Methods," Mar., 2016.
6. Invited research talk and tutorial "Method of Moments and Tensor Decomposition" at Janelia Research Campus, Ashburn, Virginia, Oct., 2015.
5. Oral presentation in COLT, "Escaping from Saddle Points — Online Stochastic Gradient for Tensor Decomposition," Paris, France, Jul., 2015.
4. Spotlight talk at Non-convex optimization NIPS Workshop, "Escaping from Saddle Points — Online Stochastic Gradient for Tensor Decomposition," Dec., 2015.
3. Spotlight talk "Method of Moments and Tensor Decomposition" at BigNeuro NIPS Workshop, Dec., 2015.
2. Contributed Talk "Fast Detection of Overlapping Communities Via Online Tensor Methods" at Topic Model NIPS Workshop, Dec., 2013.
1. Invited Talk in AI/ML seminar, Donald Bren Hall ICS department at UC Irvine, Mar., 2013.

Editorial and Peer-Review Service

14. **Program Chair, ICML:** appointed for 2027.
13. **Action Editor:** Transactions on Machine Learning Research (TMLR), Jan. 2024–present.
12. **Guest Editor:** Frontiers Research Topic "Tensor Methods for Deep Learning," 2021.
11. **Senior Area Chair, ICLR:** 2025, 2026; appointed for 2027.
10. **Senior Area Chair, ICML:** 2026.
9. **Senior Program Committee, AAAI:** 2026.
8. **Area Chair, AAAI and CoRL:** AAAI (appointed for 2027); CoRL (2026).
7. **Area Chair, NeurIPS:** 2021–2025.
6. **Area Chair, ICML:** 2019, 2020, 2025.
5. **Area Chair, ICLR:** 2021–2024.
4. **Program Committee:** AAAI (2019); IJCAI-ECAI (2018); SysML (2018, 2019).
3. **Recent conference reviewing:** ACL Rolling Review (ARR), CVPR, ECCV, and NeurIPS (2026).
2. **Additional selected conference reviewing:** AISTATS, GLOBECOM, ICLR, ICML, IJCAI, ISIT, MobiHoc, and MILCOM.
1. **Selected journal reviewing:** *Annals of Statistics*; *Journal of Machine Learning Research*; *IEEE Signal Processing Magazine*; *IEEE/ACM Transactions on Networking*.

Media and Public Engagement

21. UMD CS alumnus Ananth Sankaralingam (B.S. '26), co-founder of Y Combinator-backed Chromie.dev, discussed how CMSC 848N broadened his view of what could be built with large language models in "Learning When to Pivot," UMD Department of Computer Science, Aug. 2026.
20. Profile interview on reliable AI agents, world models, robotics, and research mentorship, "Furong Huang Builds AI Systems Designed to Act Reliably," UMD Department of Computer Science, Apr. 2026.
19. HomeGraph robotics research and NVIDIA Academic Grant support were featured by UMIACS, Apr. 2026.
18. UMD's PropensityBench framework was featured in CMNS News after Meta used it to evaluate the safety of a new multimodal AI model, Apr. 2026.
17. Research on latent safety risks in agentic AI was featured in CMNS News, Dec. 2025.
16. Research on evasive AI behavior and support from Open Philanthropy were featured by UMIACS, Nov. 2025.
15. Research on security threats to web AI agents was featured in CMNS News, May 2025.
14. Commercialization of AI-safety research was featured in CMNS News, Mar. 2025.
13. The NeurIPS Competition on Erasing the Invisible, co-organized to stress-test image watermarks, was featured by UMIACS, Jan. 2025.
12. Keynote on reinforcement learning and game theory at the New York Academy of Sciences Machine Learning Symposium was covered by the UMD Department of Computer Science, Nov. 2024.

11. DARPA-funded research on foundation models for autonomous robotics was featured by UMIACS, Sept. 2024.
10. Interviewed by *Computer Vision News* on trustworthy AI, reinforcement learning, and the future of the field, June 2024.
9. Guest on the DCCxAI podcast on trustworthy AI, bias, and deepfakes, May 2024.
8. Interviewed by the Institute for Trustworthy AI in Law & Society (TRAILS) for "TRAILS Researchers Discuss How AI Will Change the Way We Live," May 2024.
7. Hosted a public Reddit Ask Me Anything on artificial intelligence and machine learning, May 2024.
6. Participation in the federal National Artificial Intelligence Research Resource pilot was featured by UMIACS, May 2024.
5. Research on autonomous jailbreak attacks and large-language-model safety was featured by Science News Explores, Apr. 2024, and Science News, Feb. 2024.
4. Research on the limits of AI-generated text detection was named one of the CMNS Top Stories of 2023, Dec. 2023, and covered by TechXplore, May 2023.
3. Research on fairness in AI and the NSF–Amazon Fairness in AI program was featured in Maryland Today, Apr. 2022.
2. Best Paper recognition at the NeurIPS 2021 SafeRL Workshop was featured by the UMD Department of Computer Science, Jan. 2022.
1. UMD's partnership with Capital One was covered by the Washington Business Journal, Nov. 2018.

TEACHING, MENTORING, AND ADVISING ACTIVITIES

Courses Taught

16. Fall 2026, CMSC 848N, *Generative AI Agents* (course website).
15. Summer 2026, Generative AI Agent, DeepLearn 2026, 13th International School on Deep Learning.
14. Spring 2026, CMSC 422, *Introduction to Machine Learning* (course website)
13. Fall 2025, CMSC 848N, Generative AI Agents(course website)
12. Spring 2024, CMSC 422, *Introduction to Machine Learning* (course website)
11. Fall 2023, CMSC 742, *Algorithms in Machine Learning: Guarantees and Analyses* (course website)
10. Spring 2023, CMSC 742, *Algorithms in Machine Learning: Guarantees and Analyses* (course website)
9. Fall 2022, CMSC 422, *Introduction to Machine Learning* (course website)
8. Spring 2022, CMSC 422, *Introduction to Machine Learning* (course website)
7. Fall 2021, CMSC 742, *Algorithms in Machine Learning: Guarantees and Analyses* (course website)
6. Spring 2021, CMSC 422, *Introduction to Machine Learning* (course website)
5. Fall 2020, CMSC 828U, *Algorithms in Machine Learning: Guarantees and Analyses* (course website)
4. Spring 2019, CMSC 828U, *Algorithms in Machine Learning: Guarantees and Analyses* (course website)
3. Fall 2018, CMSC 498V, *Advanced Topics in Machine Learning* (course website)
2. Spring 2018, CMSC 422, *Introduction to Machine Learning* (course website)
1. Fall 2017, CMSC 828R, *Topics in Machine Learning* (course website)

Course and Curriculum Development

3. Developed CMSC848N, a graduate course on Generative AI Agents, including lectures and projects focused on LLM-based reasoning, planning, tool use, memory, and evaluation. Designed new assignments and hands-on projects on building robust agentic systems for real-world applications.
2. Developed new Machine Learning graduate course, including lectures and projects, focusing on ML methods that provide theoretical guarantees on performance. Methods taught are applicable to real-world problems, including social networks, biological systems, and natural language processing.
1. Developed new topics for undergraduate course, including Learning Theory, Latent Variable Models, and principled methods in Deep Learning. Developed new assignments and projects for these topics.

Research Advising

Doctoral Students and Research Collaborators

Formal Ph.D. advisees and research collaborators are identified explicitly below.

43. Yucheng Pan, Ph.D. advisee, 2026–present
42. Jiawei Xu, Ph.D. advisee, 2025–present
41. Andrew Mendez, Ph.D. advisee, 2025–present
40. Minghui Liu, Ph.D. advisee, 2025–present
39. Sy-Tuyen Ho, Ph.D. advisee, 2025–present
38. Weize Liu, Ph.D. advisee, 2025–present
37. Lingzhi Yuan, Ph.D. advisee, 2025–present
36. Yoonkyo Jung, Ph.D. advisee, 2025–present
35. Shayan Shabihi, Ph.D. advisee, 2025–present
34. Seungjae (Jay) Lee, Ph.D. advisee, 2024–present
33. Michael-Andrei Panaitescu-Liess, Ph.D. advisee, 2024–present
32. Cheryl Yongyuan Liang, Ph.D. advisee, 2024–present
31. Aakriti Agrawal, Ph.D. advisee, 2023–present
30. Souradip Chakraborty, Ph.D. advisee, 2022–present
29. Pankayaraj Pathmanathan, Ph.D. advisee, 2022–present
28. Chenghao Deng, Ph.D. advisee, 2021–present
27. Xiyao Wang, Ph.D. advisee, 2022–2026 (Tencent Hunyuan, North America)
26. Mucong Ding, Ph.D. advisee, 2021–present
25. Frank Ruijie Zheng, Ph.D. advisee, 2022–present
24. Sicheng Zhu, Ph.D. advisee, 2022–2025
23. Bang An, Ph.D. advisee, 2021–2025
22. Xiaoyu Liu, Ph.D. advisee, 2020–2024
21. Marco Bornstein, Ph.D. advisee, 2020–2025
20. Yuancheng Xu, Ph.D. advisee, 2020–2025
19. Yifan Yang, doctoral research mentee, 2020–2025
18. Xiangyu Liu, Ph.D. advisee, 2022–2024
17. Tahseen Rabbani, Ph.D. advisee, 2019–2024
16. Zora Che, Ph.D. advisee, 2023–2024
15. Jiaxin Yuan, doctoral research mentee, 2022–2026
14. Wichayaporn Wongkamjan, Ph.D. advisee, 2021–2024
13. Yanchao Sun, Ph.D. advisee, 2018–2023
12. Jiahao Su, Ph.D. advisee, 2017–2022
11. Zhi Zhang, Ph.D. advisee, 2020–2021
10. Chen Chen, Ph.D. advisee, 2019–2021
9. David Haoran Chan, Ph.D. advisee, 2021
8. MG Hirsch, Ph.D. advisee, 2021
7. Jin-peng Liu, research collaborator, 2018–2020
6. Tongyang Li, research collaborator, 2017–2020
5. Xuchen You, Ph.D. advisee, 2017–2019
4. Jingling Li, Ph.D. advisee, 2017–2019
3. Yogesh Balaji, research collaborator, 2017–2019
2. Seyed Abdulaziz Esmaeili, Ph.D. advisee, 2017–2018
1. Jialin Li, Ph.D. advisee, 2017–2018

Master's Research Mentoring

All entries below are M.S. research advisees.

10. Dat Dao, 2025–present
9. Xinchen Yu, 2025–present
8. Rohit Suresh, 2025–present
7. Andrew Zheng, 2025–present
6. Mohneet Kaur, 2026–present
5. Kimia Samieinejad, 2025–present
4. Ignacio Hernández Bas, 2025–present
3. Quan Nguyen, 2025–present

2. Vinay Samuel, 2025–present
1. Sudhamshu Dandu, 2025–present

Undergraduate Research Mentoring

All entries below are undergraduate research mentees.

47. Raghav Thind, 2026 - present
46. Sarvesh Baskar, 2025 - present
45. Muhammad Islam, 2026 - present
44. Kaushal Janga, 2025 - present
43. Andrew Wang, 2025 - present
42. Ankit Nakhawa, 2025 - present
41. Nandhu Pillai, 2026 - present
40. Amar Dhillon, 2026 - present
39. Jude Lwin, 2025 - present
38. Arjun Rajaram, 2025 - present
37. Aayush Talreja, 2025 - present
36. Hyunwoo Jae, 2025 - 2026
35. Keenan Powell, 2025 - 2026
34. Neel Jay, 2025 - 2026
33. Sungbin Oh, 2025 - 2026
32. Aadi Palnitkar, 2025 - 2026
31. Rohit Karthickeyan, 2025 - 2026
30. Satya Shah, 2025 - 2026
29. Anirudh Pappu, 2025 - 2026
28. Jess Choe, 2025 - 2026
27. Mingzhi Chen, 2025 - 2026
26. Shile Li, 2025 - 2026
25. Jiaming Meng, 2025 - 2026
24. Ignacio Hernandez Bas, 2025 - 2026
23. Baozan Yan, 2025 - 2026
22. Andrew Xu, 2025 - 2026
21. Anirudh Satheesh, 2023 - present
20. Justin Huang, 2022-2023
19. Benjamin Kreiswirth, 2022-2023
18. Evan Wang, 2021-2023
17. Zora Che, 2021-2023
16. Cheryl Yongyuan Liang, 2021-2023
15. Frank Ruijie Zheng, 2020-2022
14. Kate Hu, 2020-2021
13. Arjun Rajkumar, 2019-2021
12. Huimin Zeng, 2019-2021
11. Elly Do, 2021
10. George Li, 2021
9. Son Phan, 2020
8. Shiqi Wang, 2019-2020
7. Xiangyu Ying, 2019-2020
6. Zhixuan Zhou, 2019-2020
5. Amy Huddell, 2019
4. Ruochun Wang, 2019
3. Jeremy Hu, 2018- 2020
2. Christopher DeCarolis, 2017- 2019
1. Yu Wang, 2017- 2018

Junior Faculty Mentoring

2. Fumeng Yang, 2024–present.
1. Sanghamitra Dutta, 2022–present.

Ph.D. and M.S. Committee and Examination Service

Unless another institution is named, service was at the University of Maryland. UMD units: CS (Computer Science), ECE (Electrical and Computer Engineering), AMSC (Applied Mathematics, Statistics, and Scientific Computation), INFO (Information Studies), and CMNS (College of Computer, Mathematical, and Natural Sciences).

81. Ming Li (Ph.D.) — Preliminary examination committee member, CS (2026).
80. Jie Li (Ph.D.) — Preliminary examination committee member, CS (2026).
79. Jiuhai Chen (Ph.D.) — Committee member, CMNS (2026).
78. Dylan Sam (Ph.D.) — Ph.D. committee member, Carnegie Mellon University (2025–2026). Advisor: Zico Kolter.
77. Yingchen Xu (Ph.D.) — External Ph.D. examiner, UCL (2026). Advisors: Tim Rocktäschel and Edward Grefenstette.
76. Tom Sander (Ph.D.) — External thesis reviewer (rapporteur) and defense jury member, École Polytechnique / Meta FAIR (2026). Advisors: Alain Durmus and Chuan Guo.
75. Sanae Lotfi (Ph.D.) — Dissertation committee member, NYU (2025). Advisor: Andrew Gordon Wilson.
74. Hanyu Wang (Ph.D.) — Committee member, CMNS (2025).
73. Yuxin Wen (Ph.D.) — Committee member, CS (2025).
72. Faisal Adamu Hamman (Ph.D.) — Committee member, ECE (2025).
71. Shuaiyi Huang (Ph.D.) — Committee member, CS (2025).
70. Wenxiao Wang (Ph.D.) — Committee member, CS (2025).
69. Ryan Peter Sullivan (Ph.D.) — Committee member, CS (2025).
68. Samyadeep Basu (Ph.D.) — Committee member, CS (2025).
67. Fuxiao Liu (Ph.D.) — Committee member, CS (2025).
66. Dehao Yuan (Ph.D.) — Committee member, CS (2025).
65. Yuhang Zhou (Ph.D.) — Dean's representative, INFO (2025).
64. Arpit Bansal (Ph.D.) — Committee member, CS (2025).
63. Zachary McBride Lazri (Ph.D.) — Committee member, ECE (2025).
62. Gowthami Somepalli (Ph.D.) — Committee member, CS (2024).
61. Sakila S. Jayaweera Samaranayake (Ph.D.) — Committee member, ECE (2024).
60. Wei-Hsiang Wang (Ph.D.) — Committee member, ECE (2024).
59. Alireza Ganjdanesh (Ph.D.) — Committee member, CS (2024).
58. Neha Mukund Kalibhat (Ph.D.) — Committee member, CS (2024).
57. Wenhan Xian (Ph.D.) — Committee member, CS (2024).
56. Zhengmian Hu (Ph.D.) — Committee member, CS (2024).
55. Priyatham Kattakinda (Ph.D.) — Committee member, ECE (2024).
54. Aquia Richburg (Ph.D.) — Committee member, AMSC (2024).
53. Kezhi Kong (Ph.D.) — Committee member, CS (2024).
52. Ruichen Wang (Ph.D.) — Committee member, ECE (2024).
51. David Li (Ph.D.) — Committee member, CS (2024).
50. Thomas David Wrona (M.S.) — Committee member and Dean's representative, CS (2023).
49. Henok Gebeyehu (M.S.) — Committee member, ECE (2023).
48. Neha Kalibhat (Ph.D.) — Preliminary examination committee member, CS (2023).
47. Renkun Ni (Ph.D.) — Committee member, CS (2023).
46. Bo He (Ph.D.) — Committee member, CS (2023).
45. Wenhan Xian (Ph.D.) — Preliminary examination committee member, CS (2023).
44. Manli Shu (Ph.D.) — Committee member, CS (2023).
43. Shlok Kumar Mishra (Ph.D.) — Committee member, CS (2023).
42. Biao Jia (Ph.D.) — Committee member, CS (2023).
41. Alexander Jacob Levine (Ph.D.) — Committee member, CS (2023).
40. Yu Shen (Ph.D.) — Committee member, CS (2023).
39. Yigitcan Kaya (Ph.D.) — Committee member, CS (2023).
38. Samuel William Atwell Dooley (Ph.D.) — Committee member, CS (2023).
37. Valeriia Cherepanova (Ph.D.) — Committee member, AMSC (2023).
36. Brandon Feng (Ph.D.) — Committee member, CS (2023).

35. Hao Chen (Ph.D.) — Committee member, CS (2023).
34. Yushan Feng (Ph.D.) — Committee member, CS (2023).
33. Avi Koplon Schwarzschild (Ph.D.) — Committee member, AMSC (2023).
32. Jordan Kirby Terry (Ph.D.) — Committee member, CS (2023).
31. Jun Wang (Ph.D.) — Committee member, ECE (2022–2023).
30. Justin Terry (Ph.D.) — Committee member, CS (2023).
29. Mohammad Amin Ghiasi (Ph.D.) — Committee member, CS (2022).
28. Varun Suryan (Ph.D.) — Committee member, CS (2022).
27. Xin Tian (Ph.D.) — Committee member, ECE (2022).
26. Chen Zhu (Ph.D.) — Committee member, CS (2022).
25. Shouvanik Chakrabarti (Ph.D.) — Committee member, CS (2022).
24. Zeyad Ali Sami Emam (Ph.D.) — Committee member, AMSC (2022).
23. Chihuang Liu (Ph.D.) — Committee member, ECE (2021–2022).
22. Gaurav Shrivastava (M.S.) — Committee member, CS (2021).
21. Parsa Saadatpanah (Ph.D.) — Committee member, CS (2021).
20. Yuqian Hu (Ph.D.) — Committee member, ECE (2021).
19. Wells Robinson (Ph.D.) — Committee member, CS (2021).
18. Xitong Yang (Ph.D.) — Committee member, CS (2021).
17. Mingliang Chen (Ph.D.) — Committee member, ECE (2021).
16. Fengyu Wang (Ph.D.) — Committee member, ECE (2021).
15. Sushant Patkar (Ph.D.) — Committee member, CS (2021).
14. Sai Deepika Regani (Ph.D.) — Committee member and Dean's representative, ECE (2020–2021).
13. Xiaoxu Meng (Ph.D.) — Committee member, CS (2020).
12. Ali Shafahi (Ph.D.) — Committee member, CS (2020).
11. Tongyang Li (Ph.D.) — Committee member, CS (2020).
10. Yuan Su (Ph.D.) — Committee member, CS (2020).
9. Yiqi Li (M.S.) — Committee member and Dean's representative, ECE (2019).
8. Xing Niu (Ph.D.) — Committee member, CS (2019).
7. Roozbeh Yousefzadeh (Ph.D.) — Committee member, CS (2019).
6. Thang Dai Nguyen (Ph.D.) — Committee member, CS (2019).
5. Karthik Abinav Sankararaman (Ph.D.) — Committee member, CS (2019).
4. Pan Xu (Ph.D.) — Committee member, CS (2019).
3. Zheng Xu (Ph.D.) — Committee member, CS (2019).
2. Ruofei Du (Ph.D.) — Committee member, CS (2018).
1. Mohammed E. Fathy Salem (Ph.D.) — Committee member, CS (2018).

SERVICE ACTIVITIES

Professional Service

8. Co-organizer, ICLR 2025 Workshop on Bidirectional Human–AI Alignment (Bi-Align), Singapore, Apr. 2025.
7. Organizer of NeurIPS 2024 Competition: Erasing the Invisible: A Stress-Test Challenge for Image Watermarks, Sep. - Dec. 2024.
6. Chair of NSF-Amazon FAI (Fairness in AI) PI Meeting, Jan. 9-10, 2024, Arlington, VA.
5. Co-chair of NSF-IEEE workshop: Toward Explainable, Reliable, and Sustainable Machine Learning in Signal & Data Science, Mar. 20-21, 2023, College Park, MD.
4. Chair, IEEE Signal Processing Society (SPS) Washington Chapter, 2022–present.
3. Organizer: "Dagstuhl Seminar on Tensor Computations: Applications and Optimization" 2020-2021 and 2021-2022.
2. Organizer: "Matrix Factorization" Workshop at 5th Heidelberg Laureate Forum, Heidelberg, Germany, Sep. 2017.
1. Program Steering Committee, SysML 2018.

Non-University Panels

9. NSF panelist at Electrical, Communications, and Cyber Systems (ECCS) EPCN Unsolicited Panel 0515, May. 2023.
8. NSF panelist at CCF-CIF Small ML+Optim P in the Division of Computing and Communications Foundations, Apr. 2020.
7. Jury of the ACM India Doctoral Dissertation Award, Best thesis, 2020.
6. NSF panelists at RI CRII RH Panel in the Information & Intelligent Systems Division, Dec., 2019.
5. Jury of the ACM India Doctoral Dissertation Award, Best thesis, 2019.
4. NSF panelists at CISE/IIS ML/AI Medium Projects Panel, Feb., 2019.
3. Jury of the ACM India Doctoral Dissertation Award, Best thesis, 2018.
2. SysML Conference Panel, 2018.
1. US-Czech AI Workshop Panel, 2018.

University Service

32. Moderator and panelist, “Foundations: Theory, Language & Quantum,” Computer & Computational Sciences Research Day, University of Maryland, Aug. 2026.
31. Faculty panelist, Computer Science Graduate Orientation, University of Maryland, Aug. 2026.
30. Strategic Planning Committee, Department of Computer Science, 2026–2027.
29. Chair, Diversity and Inclusion Committee, 2025–2026.
28. AIM Tenure-Track Faculty Search Committee, 2025–2026.
27. UMIACS Director Search Committee, 2025–2026.
26. Merit Review Committee, 2024.
25. UMIACS APT Committee, 2023 - 2024.
24. Faculty Hiring Committee, 2023-2024.
23. University of Maryland AI Working Group Committee, University of Maryland, 2023.
22. Rising Star in ML Committee, 2023.
21. Department council, 2022-2023.
20. Graduate student fellowship committee, 2022-2023.
19. I4C Director Search Committee, 2022-2023.
18. Field Committee Chair - Artificial Intelligence, 2022-2023.
17. Rising Star in ML Committee, 2022.
16. Admission Committee, 2021-2022.
15. Field Committee Chair - Artificial Intelligence, 2021-2022.
14. Faculty Hiring Committee, 2020-2021.
13. Rising Star in ML Committee, 2021.
12. Field Committee Chair - Artificial Intelligence, 2020-2021.
11. STAT 400 CS Department Representative, 2020.
10. Rising Star in ML Committee, 2020.
9. Field Committee Chair - Artificial Intelligence, 2019-2020.
8. CS-ECE joint capstone project course design, 2019.
7. Rising Star in ML Committee, 2019.
6. AMSC SC track reform committee, 2019.
5. Academic Job Search Panel, 2019.
4. Field Committee Chair - Artificial Intelligence, 2018-2019.
3. Middlestates Assessment Committee, 2018-2019.
2. Faculty Hiring Committee, 2017-2018.
1. Committee for Google PhD Fellowship Department Nomination, Nov. 2017.

SYNERGISTIC ACTIVITIES

- **Trustworthy and agentic AI with real-world impact.** Research connects foundations of machine learning with AI safety, fairness, robustness, multimodal reasoning, and sequential decision-making. Partnerships with government, nonprofit, and industry sponsors translate these methods into deployable evaluations and systems.

- **Research communities and convening.** Organizer of the Matrix Factorization Workshop at the fifth Heidelberg Laureate Forum; Dagstuhl Seminars on Tensor Computations (2020, 2022); Rising Star in Machine Learning; and the University of Maryland Machine Learning Reading Group.
- **Scientific leadership.** Senior Area Chair for ICLR (2025, 2026; appointed for 2027) and ICML (2026); Area Chair for ICLR, ICML, NeurIPS, CoRL, and AAAI; Action Editor for TMLR; and Program Committee member for IJCAI-ECAI and SysML.
- **Broad research dissemination.** More than 80 invited, keynote, panel, and research presentations across academia, government, and industry, including the tutorial "The Role of Tensors in Deep Learning" at ACM SIGMETRICS 2018.
- **Open-source research.** Research code, datasets, benchmarks, and project pages are released through the Huang Group GitHub repository and the artifact links accompanying the publication record.